

Summarized chain of thought (Symmetric and general Mahler conjectures)

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These three attempts address complementary parts of the Mahler problem. Part I concerns the sharp symmetric lower bound. Part II takes that bound as a supplied premise and seeks the Hanner equality classification; its supplied premise and candidate reasoning are not identified with the particular proof in Part I. Part III develops a separate general-body argument for the simplex bound and its equality cases, rather than deriving it from the symmetric attempts. The three parts retain separate arguments and hypotheses.

Part I: The symmetric lower bound

Original prompt (excerpts)

Problem

For an integer $n \geq 4$, let $K \subset \mathbb{R}^n$ be a compact convex set with nonempty interior, with $K = -K$ and 0 in its interior. Define its polar by

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$K^\circ = \{y \in \mathbb{R}^n : \langle x, y \rangle \leq 1 \text{ for every } x \in K\},$
 $\langle x, y \rangle = \sum_{i=1}^n x_i y_i.$

\$\$

Writing vol_n for n -dimensional Lebesgue volume, is it true that

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$\text{vol}_n(K) \text{vol}_n(K^\circ) \geq \frac{4^n}{n!}$

\$\$

for every $n \geq 4$ and every such K ?

1 Boundary laws and the symmetric bound

The symmetric attempt begins by recalling the three-dimensional result of Iriyeh–Shibata, Meyer’s unconditional argument, and the lower bounds of Bourgain–Milman, Kuperberg, and Nazarov [36, 46, 15, 39, 49]. It tries section induction, face flags, equal-measure cone partitions, entropy, and tensorization, but finds no sharp constant along these routes. In the polytope approach, it invokes Stanley’s lower bound for the h -vector of centrally symmetric simplicial polytopes, but still needs a bound for the associated volume fractions [62]. It revisits Kuperberg’s neck construction and Berndtsson’s complex-integral proof, examines Nazarov’s Bergman-kernel approach, and considers Klartag’s relation between isotropic constants and Mahler volumes [39, 9, 49, 38].

Lundin’s extremal-function formula and the Baran metric lead it toward planar boundary laws and complexified slabs [44, 5, 6]. It chooses

$$f(\zeta) = \frac{8}{\pi^2} \sum_{k \geq 1, k \text{ odd}} \frac{\zeta^k}{k^2},$$

whose boundary real part is a triangle wave, hence uniformly distributed on $[-1, 1]$. For each row basis of a slab presentation of K , it defines a feasibility probability P_I using independent boundary values.

It uses powers of the inverse conformal map, positivity of the complex Hessian of a logarithmic potential, Stokes' theorem, and a homogeneous flux calculation to derive $\sum_I P_I \geq 1$. For the opposite inequality, it associates feasible bases with disjoint simplices in K° , obtaining $\sum_I P_I \leq n!|K||K^\circ|/4^n$. It repeatedly checks normalization, genericity, boundary ties, and Hanner examples, then presents the argument with approximation from slab polytopes to arbitrary symmetric bodies.

Part II: Equality and Hanner polytopes

Original prompt (excerpts)

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# Symmetric Mahler–Hanner: remaining equality classification

Let  $(\cdot)$  denote Lebesgue measure in  $(\mathbb{R}^n)$ , and let  $(\langle \cdot, \cdot \rangle)$  be the standard inner product. A convex body in  $(\mathbb{R}^n)$  is a compact convex set with nonempty interior. Write


$$\mathcal{K}_0^n = \{K \subset \mathbb{R}^n: K \text{ is a convex body and } K = -K\}.$$


For  $(K \in \mathcal{K}_0^n)$ , define its polar body by


$$K^\circ = \{y \in \mathbb{R}^n: \langle x, y \rangle \leq 1 \text{ for all } x \in K\},$$


and its volume product by


$$P(K) = |K| \cdot |K^\circ|.$$


For convex bodies  $(K \subset E)$  and  $(L \subset F)$  in finite-dimensional real vector spaces, define their  $(\ell_\infty)$ - and  $(\ell_1)$ -sums in  $(E \oplus F)$  by


$$\begin{aligned} K \oplus_\infty L &= K \times L, \\ K \oplus_1 L &= \operatorname{conv}((K \times \{0\}) \cup (\{0\} \times L)). \end{aligned}$$


The class of Hanner polytopes is the smallest class of origin-symmetric convex polytopes containing all origin-symmetric line segments and closed under  $(\oplus_\infty)$  and  $(\oplus_1)$ .

Open problem: For every integer  $n \geq 4$  and every origin-symmetric convex body  $K$  in  $(\mathcal{K}_0^n)$ , prove or disprove that


$$P(K) = \frac{4^n}{n!}$$


 $\iff$ 

 $K$  is linearly equivalent to an  $n$ -dimensional Hanner polytope.

A negative answer must give a mathematically valid counterexample to this equivalence. A positive answer must establish it in every dimension  $n \geq 4$ .
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2 A lens argument for Hanner equality

The attempt takes a previously supplied sharp symmetric lower bound as a premise and reconstructs an earlier candidate argument for the equality case. That argument includes a lens construction and routes from its equality conditions to the three-ball intersection property.

It first checks that polarity interchanges the two Hanner sums and that either sum in dimensions k, l

has volume product $P(A)P(B)/\binom{k+l}{k}$. Induction from segments therefore gives the Hanner direction.

For the converse, the assistant studies the analytic map

$$F(z) = \frac{8}{\pi^2} \sum_{j \geq 0} \frac{(-1)^j z^{2j+1}}{(2j+1)^2}.$$

It checks convexity of its inner image curves through the positivity of $\operatorname{Re}(1+zF''/F')$. The limiting boundary has branches $\pm\lambda(t) + it$, with uniform height under uniform phase and $\lambda''(t) = -\sec(\pi t/2)$. For spanning real rows it forms a sum $S(r)$ of basis-feasibility probabilities. Splitting a smoothed complex Hessian determinant into minors containing a constant row and those omitting it, it derives monotonicity: the latter terms decrease with r , integration by parts makes total cutoff-mass differences tend to zero, and the former terms converge to feasibility probabilities. It checks extra-row ties through a rank-two phase map and uses dominated convergence at the circle and lens endpoints.

The assistant obtains the circle value $S(0+) = 1$ by randomizing squared radii exponentially, integrating complex Gaussians, and applying Cauchy–Binet. It also considers a deterministic explanation, but finds phase choices for which two bases are feasible simultaneously:

VERBATIM EXCERPT

| Then some phase none to balance. Thus no deterministic. Laplace solution concise.

At the lens endpoint, the assistant bounds the total volume of the disjoint feasible simplices by the volume of the polar body. Integrating over the original body yields the symmetric lower bound $P(K) \geq 4^n/n!$. It then examines the equality conditions.

For an equality body the assistant lifts to $K \oplus_1 [-1, 1]$, whose polar is $Q \times [-1, 1]$, where $Q = K^\circ$. It uses inscribed product approximations and vanishing integrated missing volume to extract cost-optimal representations with minimum total mass. The asymptotic $\lambda(1 - \eta) = (2/\pi)\eta \log(1/\eta) + O(\eta)$ turns those costs into affine expressions. Comparing two representations and separating their compact convex aggregate set yields, in its reconstruction, a metric median for any three points in the norm with unit ball Q . Moving such a median toward a center whose ball misses it gives a common point of three pairwise-intersecting balls.

The earlier candidate argument also discusses a facial route using Lima’s intersection characterization [43]; the assistant chooses the median route and checks the real, finite-dimensional hypotheses of Hansen–Lima’s structure theorem, particularly Corollary 7.4 [35]. It concludes that Q is an iterated ℓ_1 - and ℓ_∞ -sum of segments and applies polarity to conclude the stated classification for K .

Part III: General convex bodies

Original prompt (excerpts)

Problem

For an integer $n \geq 2$, a convex body $K \subset \mathbb{R}^n$ is a compact convex set with nonempty interior $\operatorname{int} K$. Write $|A|$ for the n -dimensional Lebesgue volume of a measurable set $A \subset \mathbb{R}^n$ and $\langle x, y \rangle = \sum_{j=1}^n x_j y_j$ for the Euclidean inner product.

For $z \in \operatorname{int} K$, define
 $\$$

$$(K-z)^{\circ} = \{y \in \mathbb{R}^n : \langle y, x-z \rangle \leq 1 \text{ for every } x \in K\},$$

$$\text{\quad\quad}$$

$$P(K) = \inf_{z \in \operatorname{int} K} |K| \cdot |(K-z)^{\circ}|.$$

$$\$$$

The infimum is attained at a unique interior point, called the Santaló point. A simplex is the convex hull of $n+1$ points $v_0, \dots, v_n \in \mathbb{R}^n$ for which $v_1 - v_0, \dots, v_n - v_0$ are linearly independent.

Resolve the conjecture that, for every integer $n \geq 2$ and every convex body $K \subset \mathbb{R}^n$,

$$P(K) \geq \frac{(n+1)^{n+1}}{(n!)^2},$$

$$\$$$

with equality if and only if K is a simplex. The quantifier includes arbitrary convex bodies; no central symmetry, smoothness, or polytope assumption is imposed.

3 Conic reformulation and the first failed reductions

The assistant first considered whether tensor powers could turn an asymptotically sharp bound into the simplex constant, comparing this possibility with Kuperberg's symmetric lower bound [39]. It homogenized the body into a cone of dimension $d = n + 1$, and rewrote the target as

$$\chi_C(v) \chi_{C^*}(u) \geq 1, \quad \chi_C(v) = \int_C e^{-\langle v, x \rangle} dx, \quad \langle u, v \rangle = d.$$

The orthant supplied the equality model. Klartag's logarithmic Laplace formulation [38] led it to covariance restrictions at projective minima. It then noticed that smooth bodies could have interior projective minima, defeating its hoped-for universal trace estimate.

Triangulations expressed the two integrals through determinants of slack matrices. Matching opposite face flags gave equality for the orthant. The assistant then found that fine approximations to a smoothed simplex could make the matched sum small: long, nearly flat primal pieces corresponded to tiny dual pieces. A transport built from exponential quantiles encountered another obstruction. Redundant generators changed the auxiliary Cauchy–Schwarz bound without changing the cone. The assistant abandoned this estimate and examined complementary subspaces, Gale transforms, and a zonoid comparison. Although the zonoid theorem [55] suggested a route through projections of cubes, its proposed determinant comparison lost too much on round cones. It also considered the reflection-symmetry theorem of Barthe and Fradelizi [7], but observed that permutations of multidimensional blocks were not hyperplane reflections.

Induction over tangent cones produced a sum of ratios between the body's volume and truncated tangent-cone volumes. In the simple-polytope case, the assistant recognized its reformulation:

VERBATIM EXCERPT

So tangent inequality for simple B exactly Mahler. Any B can approximate simple. Not progress unless geometry helps.

It next compared the deficits in direct and reverse Brascamp–Lieb inequalities, developing Barthe's transport argument [8] into determinant and heat-flow formulas. Separate bounds on the two integrals did not give it the required bound on their product. Dirichlet representations likewise concentrated too strongly at the center of round sections.

The assistant returned to a maximal-simplex position, recalling local minimality at the simplex [37], and placed the cone above the positive orthant. It tried to connect the resulting lower sets with the entropy identity for anti-blocking corners [22]. Saint-Raymond's inequality and Meyer's coordinate-induction method [59, 46] then suggested repeated-coordinate lifts of a convex corner. The primal

weighted integral approached the desired cone integral, but the dual weights concentrated at one point of the exposed face. The assistant ended by asking whether varying the replication proportions could recover the whole dual section.

4 Localization losses and stationary incidence measures

The assistant continued trying to transfer anti-blocking inequalities to the conic target $\chi_C(v)\chi_{C^*}(u) \geq 1$. Repeating nearly radial coordinates led it to incompatible scales: broad primal fluctuations sampled the tangent cone, while the dual integral concentrated near one point of its section B . After computing the two asymptotics, it wrote:

VERBATIM EXCERPT

| Need $|B|$ factor, not obtained.

It also tested translated corners, power transformations, and permutation chambers. Consulting Artstein-Avidan, Sadovalsky, and Sanyal's treatment of anti-blocking bodies relative to cones, it found definitions and decompositions but no direct bound for its proposed reduction [1]. Tensor amplification was another possibility: a sufficiently strong volume comparison could magnify a deficit until it contradicted a reverse Santaló bound. The assistant found determinant losses in its column-slicing estimate and questioned whether tensoring preserved the required centers, while keeping Kuperberg's bound as a possible comparison [39].

Next it used the projective covariance conditions associated with Klartag's approach [38]. Normalizing one covariance to the identity placed two cone sections inside a unit ball. The assistant investigated whether an exponential marginal at a contact point would force a ray factor and permit induction. First variations then led it to propose vertex–facet incidence measures with facet centroids as conditional means, cross-moment identities, and a stochastic matrix with eigenvalues $1, -1/n, 0$. It found that its determinant estimate lost too much to reach the target.

Section–projection induction brought a different obstruction: simplex sections chosen by a natural partition could have the wrong Santaló center. The assistant instead considered unequal orthant partitions and tracked their section and projection weights. It also developed an entropy projection onto a subset of the probability simplex, then noticed that projecting exterior points onto boundary faces made the proposed change of variables degenerate.

Reading Chen, Li, Xi, and Xu's shadow-flow argument and its use of Meyer–Reisner polar-volume convexity [19, 47], the assistant tried to extend the count of admissible vertex speeds. Higher-dimensional incidence counts, including the 24-cell, led it to reconsider universal elimination by deformation:

VERBATIM EXCERPT

| Need lower bound for rigid cases, not classification.

It returned to stationary incidence measures, interpreting them through random facet triangulations and random choices of an apex. These constructions supplied further covariance and determinant identities. It then tried to extend vertex–facet transport to complete dual face flags and noticed that artificial faces introduced by triangulation had no corresponding dual faces.

5 Barycentric coupling and the covariance gap

The assistant rewrote the face-flag estimate as a volume factor times the affinity of two flag distributions. It observed that vertex–facet stationarity did not specify transitions through intermediate faces. Returning to the covariance comparison associated with Klartag's projective variations [38], it set $z = x \cdot y$ for

independent cone-boundary points and $r = q(x) \cdot p(y)$ for their normalized normals. Boundary integration gave $\mathbb{E}[rz] = 1/n$; the desired estimate was $\mathbb{E}[z^2] \leq 1/n$. A barycentric construction produced another variable with the law of r and conditional mean z , but the assistant rejected a deduction from scalar convex order alone:

VERBATIM EXCERPT

| So no scalar alone. Need coupling constraints relating r and r' and z conditional facets.

It then proposed replacing vertices by small clusters and separating two convex sets of incidence measures to obtain one coupling satisfying both barycentric requirements. It explicitly distinguished unrestricted minimization from minimization with a fixed vertex count, and worked through approximate-contact constraints, limiting derivatives, and measurable selection for a continuum extension.

With that coupling assumed, the assistant examined reversible incidence walks, added-vertex moment bounds, and conditional determinant–entropy estimates obtained from Barthe’s reverse Brascamp–Lieb inequality [8]. It found losses for rounded facets. Meyer–Reisner shadow systems [47] motivated attempts to flatten vertices, replace bodies by pyramids, or use perspective sections; the assistant kept encountering changes in facet structure or constants below its target. A proposed common-random-vector factorization failed its regular-simplex test. Tangent degenerations led to reciprocal slack moments, while the assistant noted that Jensen’s inequality had the opposite direction from the estimate it wanted.

Further tests used regular polygons, the Birkhoff polytope, the 24-cell, and sections of positive-semidefinite cones. The assistant combined the zero-curvature condition for minimizers [56] with its proposed coupling to investigate supporting faces of positive dimension. It also considered pruning long vertices using Paouris’s concentration bound [51], and using local simplex stability [37] to separate geometries near the simplex. Tensor amplification invoked the exponential lower bound of Bourgain–Milman [15]. Symmetric suspensions and difference bodies brought in Kuperberg’s linking-integral bound and Rogers–Shephard’s inequality [39, 57]; the assistant calculated that the suspension comparison still lost an exponential factor on the simplex. It considered reflection-group lifts [7] and revisited the anti-blocking arguments of Saint-Raymond and Meyer [59, 46], tracking the opposing concentration effects of coordinate replication.

The incidence coupling next yielded an oblique projection on functions of contact pairs: its kernel was $nq \cdot p'$, and the complementary stochastic kernel was $1 - q \cdot p'$. The assistant identified a relation between a simplex projection and a simplex section. It explored a connection with Ball’s reverse isoperimetric argument [2], then tried Dirichlet barycenters with concentration invariant under replication. The concentration giving the uniform simplex was too large for the ball model. Finally, it expressed the covariance comparison through conditional-expectation operators and a Hilbert–Schmidt inequality, observing that the remaining inner product had no sign supplied by the preceding identities.

6 Symmetric lifts and the negative contact graph

The assistant expressed the incidence coupling through a conditional-expectation operator S and a non-negative slack kernel L . It obtained $K = SLS$, with zero diagonal and prescribed eigenvalues for LS , then tested the desired Hilbert–Schmidt comparison. For a permutation coupling, it found that the comparison required an additional symmetry. Nonlinear weighted-centroid maps had derivative $-nC_Y$; their composition suggested extra fixed points, but the assistant could not connect those points to a decrease in volume product. It also developed a separation argument for simultaneous barycentric transport on contact pairs and revisited the projective covariance condition from Klartag’s work [38].

The next calculations sought to amplify cone-volume bounds through tensor powers and symmetric

lifts. The assistant revisited reverse Brascamp–Lieb transport, anti-blocking corners, and shadow variations [8, 59, 46, 1, 47, 19]. It examined Kuperberg’s positive and negative contact graphs and bottleneck construction [39, 40]. For asymmetric bodies, the negative graph entered $K \times (-K^\circ)$, and fitting it into $K \times K^\circ$ introduced sign or volume losses. Order intervals, difference bodies, and permutation-invariant lifts were tested against the simplex. In considering reflection symmetries [7], it found that permutations of whole vector blocks did not provide the scalar reflections it wanted.

Zonoids offered another route through their volume-product inequality; the assistant recalled Gordon–Meyer–Reisner’s proof while remaining uncertain about its method [31]. The assistant identified a natural zonoid with a normalized projection body, but noticed that Zhang’s reverse Petty inequality gave the opposite direction from the estimate it needed [67]. It realized arbitrary polyhedral cones as tangent cones at zonotope vertices and varied generators projectively while keeping denominators fixed. The attempted induction then returned to the original cone problem:

VERBATIM EXCERPT

Realize proving global minima for projective zonoid by reducing to boundary leads tangent arbitrary unknown, no induction.

The assistant next paired the positive subgradient graph of $g = p_K^2/2$ with a flat negative graph over $K \cap (-K^\circ)$. It calculated equality in the normalized regular-simplex model, but did not establish a general comparison.

This led it to seek a convex, degree-two control potential satisfying

$$h_K(-\nabla\psi(x)) = p_K(x),$$

so that the negative gradient graph would lie in the required polar product.

VERBATIM EXCERPT

This is promising universal negative conical graph with no shrinking, unlike Kuperberg.

It examined optimal-control, subgradient, Legendre-duality, and Hamiltonian descriptions of this equation, and sought a lower bound for the join through mixed Hessian determinants. A complex Gaussian contour identity suggested retaining oscillatory cancellation instead of discarding it. The reasoning then turned to the special case $K = -K^\circ$ and to self-dual cones.

7 Negative-contact tests and stochastic entropy methods

The assistant tested a covariance trace bound for centered negatively self-polar bodies, against the projective covariance conditions in Klartag’s approach [38]. Its spherical expansion gave a positive second-order change for degree-three harmonics. It therefore rejected its expected sign, while leaving realization of the smooth centered perturbations as a further question.

Returning to Kuperberg’s linking-integral approach [39], it studied the eikonal potential ψ , with $h_K(-\nabla\psi) = p_K$, and a mixed-Hessian integral intended to bound the volume product. A Gaussian change of variables left a determinant integral that it found too small for a simplex and potentially zero for symmetric polytopes. Expanding ψ near the ball suggested evenness through third order. It then derived a polygonal compatibility condition that it thought would fail generically, and sought an explicit polygon to test it. Changing the complex contour corrected the Gaussian weight but also changed the determinant statistic; its simplex calculation gave an exponential gap between the two statistics. It also compared the graph-volume loss with a Rogers–Shephard difference-body estimate in the simplex case [57].

The assistant next consulted transport–entropy formulations of inverse Santaló inequalities [32, 26] and rederived a target involving differential entropies and maximal score correlation. For densities proportional to e^{-V} and e^{-W} , it sought

$$h(X) + h(Y) + \sup_{\pi} \mathbb{E}_{\pi}[\nabla V(X) \cdot \nabla W(Y)] \geq 3d.$$

One-dimensional slicing introduced an infimum over profiles whose integral it could not control sharply. Gaussian observations produced score martingales with increment matrices $I - t \text{Cov}(X | Z_t)$. Reparametrizing their paths appeared to give a constant larger than the target, until it noticed an endpoint problem:

VERBATIM EXCERPT

That would even stronger than Mahler (contradiction extremal). Issue endpoints r_{∞} random unequal, can't align log scales pathwise causally. Explore.

Its matrix version likewise exhausted one eigendirection before accounting for the full entropy endpoint.

After considering entropic transport, the assistant returned to coordinate couplings.

Opposite triangular transports canceled the unwanted terms for Gaussian laws, but left cross terms of uncontrolled sign for general densities. It revisited Barthe's transport proof of reverse Brascamp–Lieb [8], finding that the separate direct and reverse estimates still did not compare the required product. Further stochastic calculations used conditional entropy, covariance, and third moments in a Bellman candidate. Allowing adaptive observation speeds and either sign of innovation correlation led it to the scalar feasibility condition $|1 - G_X G_Y| \geq D_X D_Y$ when both squared quantities D_X^2, D_Y^2 were nonnegative.

8 Gaussian regularization and failed tensor amplification

The assistant found that its scalar stochastic-localization controls did not settle the terminal comparison: conditioning an exponential law reduced entropy, keeping the proposed stopping ratio above one. Replacing conditional entropy by the maximum entropy at the current mean repaired the scalar expression, but radial observations preserved the problematic ratio and left angular information unresolved. It considered Poisson observations and entropy monotonicity on upper convex sets, using Berwald's moment inequality and questioning whether Caffarelli's contraction principle gave the needed direction of comparison [12, 17].

It then introduced the centered Gaussian barrier

$$F_Q(a) = \max_m \log \int_C \exp\left(-a \cdot x - \frac{1}{2}(x - m)^T Q(x - m)\right) dx.$$

The optimizing center was the posterior mean. Differentiation gave an inflated covariance Hessian, $(\Sigma^{-1} - Q)^{-1}$, suggesting that an integrated covariance comparison might replace the sharp slicing estimate associated with Klartag's approach [38]. Unrestricted minimization let the scales escape; fixed metrics also made the balanced orthant a maximum along some opposing coordinate variations. The assistant therefore optimized over mutually inverse metrics. A strengthened Poincaré calculation gave it a covariance correction and a derivation of geodesic concavity. It used this concavity to argue that the optimized profile tensorized for cone products.

Differentiating through the metric optimum introduced an inverse Hessian correction involving third and fourth moments. Half-line computations matched the desired orthant profile:

VERBATIM EXCERPT

| Exact equality $E=p/r$ all scalar (due scaling invariance).

The assistant sought a corresponding matrix or trace estimate, developed an inductive boundary-degeneration argument, and explored Ehrhard's inequality [24]. It found that the resulting bounds and repeated Bochner estimates were too weak, particularly when covariance was small. It connected the missing estimate to sharp third-moment bounds and continued testing possible scalar extremizers.

Returning to amplification, it observed that direct cone products preserved the normalized deficit. Projective tensor cones offered a different construction, but ellipsoid estimates based on reverse Brascamp–Lieb [8] did not control the injective dual with the required constants. The assistant then reconsidered symmetry and zonoid comparisons [39, 7, 55], noting that permutations of multidimensional blocks were not hyperplane reflections. It revisited entropy and anti-blocking lifts [22, 1, 59, 46]. Repeated coordinate constraints could retain a fixed neighborhood and so failed to force the desired amplification. Localizing coordinate induction near a vertex, meanwhile, lost the matching dual projections; the assistant looked for additional coordinates that would preserve them.

9 Tensor circuits and configuration bodies

The assistant applied simplicial matrices along tensor-array coordinates, obtaining determinant sums with the desired repeated volume factors. It then found that convexifying the circuit outputs allowed adaptive conditional distributions whose volume lost the intended amplification. Arbitrary diagonal weights also broke positivity between paired primal and dual circuits. This prevented its proposed application of the Bourgain–Milman bound to the circuit sets [15].

It next sought a stationary distribution for positive slack-matrix dynamics, hoping to compare branch entropy with logarithmic determinants. The assistant identified the missing absolute continuity assumption:

VERBATIM EXCERPT

| Need avoid circularity (ac impossible if $\alpha < 1$).

Unnecessary subdivision of a simplex already made its proposed stationary-law argument fail at equality. Tensor arrays gave independent inputs to individual fibers and bounded scalar moments, but the assistant could not infer a lower bound on joint entropy. It considered entropy-power estimates for convex measures [13], while observing that mixtures after a circuit layer need not remain log-concave.

The next construction fixed the sum of cone points:

$$S_m = \{(x_1, \dots, x_m) \in C^m : \sum_i x_i = mu\}.$$

After translation to the zero-sum subspace, the assistant described its polar as a projection of a truncated product of dual cones. It used the Rogers–Shephard projection–section inequality [58] to bound the projection loss polynomially in m , with the cone dimension fixed. Decomposing the points among generating rays expressed S_m as a union of Minkowski sums of aligned simplices. Optimizing over this finite-dimensional family led it to seek an asymptotic Mahler estimate for those sums.

The assistant compared generator-length variations with Reisner's zonoid argument and Meyer–Reisner shadow systems [55, 47]. Entropy optimization showed why independent exponential lifts were too concentrated for diffuse generators; correlated lifts were needed. Direct and reverse Brascamp–Lieb comparisons [8] again left the product of their two factors uncontrolled. Returning to a functional shadow formulation, it recognized the same restriction on moving vertices:

VERBATIM EXCERPT

To move a vertex maintaining existing facets, need shadow admissibility across nonsimplicial facets as in Mahler. Fat obstruction dimension $d \geq 5$. Functional formulation not escaping.

Binary splitting produced centrally symmetric fibers, prompting comparisons with Kuperberg's bound and Saint-Raymond's unconditional inequality [39, 59]; the assistant found that its paired fibers were not polars. A subsequent basis-entropy formulation related facet densities to basis-inclusion probabilities. It considered random box sections and Vaaler's cube-section theorem [65], but noted the difference between central sections and the sections through a vertex arising here. Finally, exponential tilts of Cartesian powers gave it the covariance condition $(n + 1)(n + 2) \text{Cov}_K \text{Cov}_{K^\circ} \geq I$ at the corresponding formal minimum; it observed that the improvement over its earlier condition became small in high dimension.

10 Exponential face decompositions and complex analysis

The assistant decomposed an exponential cone law by subtracting the minimum normalized slack. The minimum had an exponential distribution independent of the boundary residual:

VERBATIM EXCERPT

Exactly memoryless along v ! Because all $s_i v = 1$ and $D + t v$. Great.

It first coupled the facet label to a triangulation, then chose a full apex triangulation separately for each label to remove information loss. Lower-dimensional induction reduced its entropy comparison to bounding average projection losses by the entropy of the facet probabilities. It recognized that this comparison returned to the original Mahler inequality. Conditional facet hazards gave a reversible graph identity; threshold variations suggested bounds on vertex degrees, but it noticed that changing triangulations produced upward kinks. Deleting a vertex instead related the degree to the volume of the removed corner.

The assistant next represented points by exponentially filtered trajectories of a Markov chain on vertices. The simplex produced the uniform law. Covariance calculations supported a conditional-barycenter identity, but higher moments introduced resolvents for which it could not justify that identity. Centroid bounds gave transition rates at least one along every edge. It also tried Lawrence's vertex-volume formula, where it encountered cancellation between signed terms [41]. Revisiting the shadow-flow arguments of Chen, Li, Xi, and Xu and Meyer-Reisner, it returned to the difficulty of deforming nonsimple minimizers [19, 47].

Turning to Nazarov's complex-analytic approach [49], the assistant proposed

$$J_p(v) = \int_C e^{-(p+1)v \cdot x} \chi_{C^*}(x)^{-p} dx.$$

An orthant-sharp upper bound for these moments would recover the desired inequality as $p \rightarrow \infty$. It tested convolution and derivative arguments; a square-cone calculation gave a negative third derivative, contradicting its proposed higher-derivative positivity. In the inverse-polar moment calculation, it applied Berwald's normalized-moment comparison to a concave geometric mean, but found that treating dual cells separately lost their shared constraint [12]. Fourier sections and Vaaler's cube-section inequality led it to seek a sharper dependence on the generating directions [65].

The assistant then tried Hardy-space functions built from facet denominators and found excessive concentration for round cones. Herglotz representations led it to a complex moment body $B_C(u)$, retaining simultaneous decompositions of u into cone vectors. Using the multidimensional Suita comparison

as a starting point [50], it sought the two bounds

$$\chi_{C^*}(u)^2 |B_C(u)| \geq \pi^d, \quad |B_C(u)| \leq \pi^d \chi_C(v)^2.$$

Its comparisons revisited Kuperberg's bound, Reisner's zonoids, Barthe–Frédérizi symmetries, Saint-Raymond's inequality, and Barthe's reverse Brascamp–Lieb theorem [39, 55, 7, 59, 8], alongside Klartag's covariance formulation [38]. Higher-rank moment bodies reduced to determinant optimization, prompting the assistant's assessment:

VERBATIM EXCERPT

| Mahler target becomes compare χ products to m_B/m ratios (taut).

It next considered Laplace operators and Gurvits's capacity inequalities [34]. Returning to slack matrices, it observed that permanent estimates did not directly control the signed determinants in the volume formulas.

11 Complex transport and weighted zonotope bounds

The assistant sought an upper volume bound for the complex moment body $B(C, u) = \{\sum_j x_j h_j : x_j \in C, \sum_j x_j = u, |h_j| \leq 1\}$. It compared this problem with Gurvits's capacity inequalities, higher-order difference bodies, and Barthe's transport proof of reverse Brascamp–Lieb [34, 60, 8]. On a simplicial piece, transporting each disk to a positive quadrant gave the desired Jacobian estimate; the assistant could not make the images from different bases disjoint or control the regions where several bases tied. Holomorphic disk maps reproduced the scalar calculation, but their first derivative alone lost a constant and a global choice of holomorphic lifts remained missing.

It next adapted the Kuperberg–Berndtsson contour argument [39, 9] to the complementarity graph $\Gamma = \{(x, y) \in C \times C^* : x \cdot y = 0\}$. Pairing this graph with a positive linear graph gave an orthant-exact lower bound involving an intersection cone and determinants on dual faces. Summing two complementarity graphs kept both outputs in their cones but introduced multiplicity. The assistant tried rank-one penalties and different complex phases, finding that discarding the oscillatory terms still lost the orthant constant.

In parallel, it considered Vaaler's section theorem [65] and a toric formulation: one cone integral became the volume of a Reinhardt domain, while the other became a fraction of integrable monomials. It compared this formulation with the multiplicity–log-canonical-threshold inequality and strong openness [25, 33]. It rejected unrestricted subadditivity by splitting a simplex into pieces, then considered restricted subdivisions, polynomial maps, and Bergman traces. A projected Cauchy-to-exponential transport also failed its proposed pointwise Jacobian test, first near the axis for symmetric examples and then along a two-level direction of a square cone.

The assistant formed an overlap kernel with Cauchy marginals in dual-cone directions. Its negative logarithm f was even and convex. It compared the homogeneous case with Schoenberg's positive-definiteness criterion and the zonoid volume-product bound [61, 55], but observed that infinite divisibility of this particular kernel would force order intervals to be zonoids. After considering Bergman-kernel subharmonicity [10], it tested holomorphic contraction of Laplace mean maps on a square cone and concluded:

VERBATIM EXCERPT

| Thus no holomorphic contraction on any bounded fixed domain in general. Need exploit station (global volume vary body). Hess alone not enough.

For a polyhedral cone $D = C^* = \{y : a_i \cdot y \geq 0\}$, it then expressed f^* through the zonotopes $Z_c = \sum_i [-c_i a_i, c_i a_i]$, where $c_i \geq 0$ and $\sum_i c_i a_i = u$. The cost assigned to each coefficient vector came from maximizing an exponentially weighted section over translated inequalities. With $J = 2^{-d} \int e^{-f^*}$, the assistant sought $1/\chi_D(u) \leq J \leq \chi_C(v)$. It noticed that arbitrary unions of the zonotopes could encode bipyramids over arbitrary symmetric bodies:

VERBATIM EXCERPT

| So pure union Mahler as hard as symmetric general!

It therefore retained the section-derived costs. Its Lorentz-cone asymptotics gave $(J/\chi_C(v))^{1/d} \sim e/4$ and $(J\chi_D(u))^{1/d} \sim \pi/2$. For the upper estimate it finally wrote an entropy identity for the density proportional to $e^{-f^*(2w-u)}$ on $[0, u]_C$, reducing the desired inequality to a lower bound of $d(1 - \log 2)$ on the sum of its mean cost and its relative entropy against the cone exponential law.

12 Facet hazards, eikonal flow, and phase losses

The assistant sought an induction for the overlap potential of an exponential cone law. On a facet hyperplane it obtained $c_i(s)e^{-f(s)} = c_i(0)e^{-f_i(s)}$, identifying the hazard-weighted restriction with a lower-dimensional overlap potential. It compared this with Meyer's divergence argument and the zonoid case [46, 55]. Kinks supplied intervals of subgradients and polar-fiber volume, but the resulting estimate required averages of squared hazards at tilted optimizers. The assistant could not obtain these averages from offset variations: the proposed stationarity condition concerned a larger family than the original cone problem. It also considered reciprocal section–projection estimates inspired by Vaaler [65].

Returning to the eikonal potential ψ , with $g = p_K^2/2$, the assistant corrected its comparison of joins. If E_k denotes an integrated mixed-Hessian coefficient, the ordinary join uses $\sum_k E_k$, whereas the conic construction weights coefficients by $1/\binom{n}{k}$. Simplex and ball calculations distinguished the two expressions. It pursued Kuperberg's contour construction and Berndtsson's complex-integral formulation [39, 9], seeking compensation between the contour's ψ -weight, the desired g -weight, and oscillatory cancellation. It also used a Golden–Thompson heat-trace bound to reduce another proposed volume estimate to controlling a ground-state energy [29, 63].

Cayley coordinates converted the gradient graph into a homogeneous field with Hessian eigenvalues in $[-1, 1]$. Gaussian determinant integrals represented the primal and polar volumes. Scaling this field toward zero gave a local second-variation estimate; the assistant found that nearly singular directions prevented its extension to the endpoints. It examined Brascamp–Lieb variance bounds [16] and recalled Klartag's covariance conditions [38]. For the simplex, it expressed the field by projecting coordinatewise absolute values onto the zero-sum hyperplane, exposing the competition between radial expansion and collapsing determinants.

The assistant next differentiated along the eikonal characteristic field. It obtained Riccati-type identities for $D^2\psi$, relating radial growth, divergence, and weighted curvature. A proposed contour deformation then reproduced the same graph under a new parametrization. Simplex asymptotics led it to insist on recovering the cancellation lost by taking absolute values.

An elliptic Hamiltonian gave a quarter-turn map from the vertical plane to the horizontal plane through Legendre duality. The assistant found that the accompanying action and billiard-length estimates fell below its target scale. For self-dual cones it tried orthogonal decompositions, but iteration required additional structure on faces; a proposed Moreau-map multiplicity bound failed for an odd-polygon cone. Its volume comparisons using Rogers–Shephard, Ball's reverse isoperimetric inequality, and Barthe's reverse Brascamp–Lieb inequality retained losses it could not remove [57, 2, 8]. It also revisited anti-blocking and Bergman-kernel routes through Saint-Raymond and the multidimensional Suita inequality [59, 50].

Finally, rare large scalar matrices contradicted an unrestricted reverse Cauchy–Schwarz inequality for averaged mixed determinants:

VERBATIM EXCERPT

| Fails! Hess homogeneous distributions may not allow scalar mass jump. Geometric path prevents.

The assistant retained the possibility of a geometrically constrained version and turned to a face-pair formula: orthogonal tangent spaces of polar faces made their contributions to the volume of $(\lambda I + \partial g)(K)$ explicit.

13 Transport moments and a weaker constant

After considering Kuperberg’s contour argument and Saint-Raymond’s unconditional result, the assistant expanded the eikonal potential around a quadratic gauge [39, 59]. For the odd perturbation $h = r^2 \cos(3\theta)$, it obtained a nonzero fifth-order odd term and used subsolutions, supersolutions, and characteristic decay to compare the expansion with the control value. It therefore rejected its proposed evenness of the potential. In the anti-polar case, it applied a moment estimate attributed to Berwald, with a binomial factor, but found that its high-moment bound lost the constant scale it needed [12]. It then reconsidered transport–entropy approaches [32], seeking a replacement for the degenerating eikonal Hessian.

The replacement was uniform Brenier transport from K to $-Q$, where $Q = K^\circ$. Its constant Jacobian led the assistant to write:

VERBATIM EXCERPT

| Great! We can choose **any** monotone map $\backslash(T\backslash) K \mapsto Q$, not eikonal, no homogeneity.

Combining this negative graph with the positive gradient graph, it derived caps on mixed-Hessian moments. It then selected a balanced linear position with $|K| = |Q| = p\kappa_n$, writing $c_k \leq \text{poly}(n) \binom{n}{k}^2$. Ball inclusion gave only the scale $p^{1/n} \geq 1/2$. For $K = -Q$, it also used Rogers–Shephard on the inclusion of its positive-gradient image in $K - K$ to bound that image’s volume [57]. A uniform-direction gradient argument also failed its simplex test: the assistant found that the volume depended on rare directions missed by that coupling.

Weighted angular integration produced recurrences for negative support moments. The assistant found that nonnegative kernels could not propagate the recurrence through the lower half of the exponent range. Projection estimates, spherical neighborhoods, and exponential kernels then led it back to curvature moments. With $h = h_Q$, $B_h = \nabla_S^2 h + hI$, and normalized elementary symmetric functions $\bar{\sigma}_k$, it wrote

$$Z_k = \mathbb{E} h^{k-n} \bar{\sigma}_k(B_h), \quad Z_k \geq \frac{w_k^{n-k+1}}{w_{k+1}^{n-k}}, \quad Z_k \leq \text{poly}(n) \binom{n}{k} p,$$

where w_k denotes the corresponding normalized intrinsic volume. Telescoping the middle ratios, it concluded that $p^{1/n} \geq e^{-1/2} + o(1)$, and checked the coefficient normalization, metric selection, and transport regularity. It compared this with its target $\sqrt{e/(2\pi)}$, identifying an accumulated logarithmic gap of about $0.081n^2$.

A conic calculation with shifted Gaussian face integrals returned the same constant. The assistant sought the missing contribution in Hölder slack, but found that abstract curvature-measure recurrences allowed sharply concentrated moving distributions. It introduced joint primal–dual face measures and

tracked the support gradients lost when passing to adjacent faces. Signed perpendicular heights prevented an immediate positive coupling of complete face chains. Finally, it used the angular partition generated by $\lambda Y + X$, for paired polar-face points Y, X . Its hypothetical zero-slack profile supplied too little angular volume. It continued by expressing the angular integral through joint negative moments of $|Y|^2$ and $|X|^2$, looking for simultaneous lower-tail contributions.

14 Centered orthoschemes and the remaining general-case obstruction

The assistant compared angular saddle constraints with the regular-simplex profile, then sought a sharp estimate for individual orthoschemes. For decreasing squared face-foot lengths A_i , it wrote their Gaussian cone integral as

$$I(A) = \int_{r_i \geq 0} \exp\left(-\frac{1}{2} \sum_{i,j} A_{\max(i,j)} r_i r_j\right) dr.$$

It applied Prékopa's marginal log-concavity theorem in logarithmic coordinates [54]. At $A_i = (n-i)/(i+1)$, Gaussian order statistics represented the logarithmic derivatives through ordered beta distributions. Monotonicity of the two Gaussian mean-excess functions supplied the desired derivative ordering:

VERBATIM EXCERPT

| Thus H decreasing. Nice. Beta stoch increasing. g monotonic all n exact! Let's note.

The assistant concluded that the resulting flag estimate could be summed for perfectly centered polytopes. It then identified two obstacles to its intended application: general flags carry signs, and the metric furnishing the transport bounds need not make the face feet lie inside their faces. A calibration using adjacent-face recurrences led it to incompatible equations on overlapping simplex regions. Replacing face feet by nearest points restored positive subdivisions but introduced zero transition heights and mismatched flag weights.

The next approach used Gaussian Moreau projection paths. Integrating projections along shifted Gaussian lines produced convex transport maps into the two cones, in the spirit of Barthe's transport proof [8]. A determinant comparison reduced the proposed estimate to a scalar rank-profile inequality. It explored arbitrary translations, convexified cost profiles, and signed walks through face subspaces. These walks retained positive quadratic energy formulas, but thin angular regions defeated the proposed local comparison. Gaussian distance profiles and the Wills functional provided further constraints; the assistant found that the resulting estimates still lost the sharp constant. It also revisited the Kuperberg bound and Berndtsson's complex-integral approach while comparing Gaussian regularization and conic contour constructions [39, 9].

The next construction transported exponential measures directly. Writing $\alpha = \chi_C(v)\chi_D(u)$, $M = DT$, and $\Delta = u \cdot T - v \cdot t$, the assistant derived the candidate averaged bound

$$\mathbb{E}[Z_M(p)e^{(1/2-p)\Delta}] \leq \sqrt{\alpha}.$$

Since $\mathbb{E}\Delta = 0$, it sought simultaneous bounds from typical products of transport Hessians. It proposed aligning the resulting axes by a change confined to a subspace of small dimension, while recording the need to control integrability, uniformity in p , and the effect on face determinants. A tilted face-balance identity then gave the expected projected energy in the aligned position. The assistant still needed nested face distributions and control of rare contributions to the angular integrals.

Finally it varied a weighted curvature functional. At a hypothetical minimizer, it interpreted its signed first-variation measure through a martingale coupling between incident faces:

VERBATIM EXCERPT

| So station imposes mass strictly flows from negative faces F to positive cofaces H !

From this proposed stationarity condition it inferred conditional bounds on vertex norms and facet distances, then began translating the signed face masses into conic notation.

15 Slicing profiles and the missing sharp constant

The assistant tried to use stationarity to balance the face coefficients of an amplified cone. Minimizing their geometric mean appeared to force the signed first-variation masses to vanish rank by rank. It then checked the proposed lower bound against the ball and changed the objective back to the largest normalized coefficient.

VERBATIM EXCERPT

| Ball violates desired geometric mean inequality!

It next expressed exterior-power coefficients through complementary projections of primal–dual contact pairs. For a fiber Q_b , its support function $h(s, b)$ gave the tangent intercept

$$\tau(s, b) = h(s, b) - b \cdot \nabla_b h(s, b).$$

The assistant interpreted this intercept through a linear program, then enlarged the slicing subspace to obtain integrals involving Hessian minors. Centering each fiber at its Santaló point suggested a concave scalar profile W , with a lower estimate involving $(W - b \cdot \nabla W)^{-k}$. Rogers–Shephard projection–section bounds and Berwald’s inequality entered the comparison with fiber volumes [58, 12]. A radial profile flat near the origin and proportional to $-r \log r$ farther out obstructed the proposed entropy-scale constant. Combining the weaker estimate with projection bounds did not yield the sought contradiction. One-coordinate recursions also lost factors, and mixed derivatives obstructed transferring the scalar calculation to general fibers.

The assistant returned to a directional criterion of the form $f_v(0)e_v(v) \leq 1/e$, pairing a central marginal density with a positive-part moment of a polar normal measure. It examined covariance restrictions associated with Klartag’s approach and a dimension-independent lower rate supplied by Bourgain–Milman [38, 15]. Meyer-style hyperplane partitions led it back to weighted versions of the same directional criterion, while Zhang’s reverse projection inequality left a dimension-dependent loss [46, 67]. Cuts through product cones did not strengthen a single cut. Adding points from complementary affine slices to contact pairs then produced a directional upper bound for an exterior coefficient; inserting lower-dimensional volume-product estimates again lost a binomial factor.

For prismatoids, the assistant wrote the fibers as $(1 - q)A + qB$ and compared their mixed-volume distribution with a binomial law. Homothetic summands gave the desired scale, whereas complementary lower-dimensional summands concentrated the distribution and lost a square-root factor. It sought compensation for this loss in the volume product of $L = C + D$, considering mixed subdivisions and variations of the summands. Looking at Reisner’s zonoid method, Meyer–Reisner shadow systems, and Saint-Raymond’s antiblocking inequality, it made further attempts to reduce sums of aligned simplices; it also revisited reverse Brascamp–Lieb comparisons [55, 47, 59, 8]. The assistant also revisited Kuperberg’s symmetric bound and Barthe–Frédérizi’s reflection-symmetry method, asking whether repeated nonsymmetric factors could acquire enough symmetry without losing the volume-product information [39, 7]. Finally, slicing a cone along an extreme ray produced an ordinary convex body with a distinguished vertex; the assistant recognized the resulting inequality as the original Mahler problem

in that description.

16 Flags, Gale densities, and degenerating decompositions

The assistant tried to combine tensor induction with projection estimates, but found that conditioning produced truncated exponentials or nonlinear gauges outside the proposed induction class. Adding a Euclidean ball through a support-function interpolation recovered the mixed-volume constraint it was trying to strengthen. Its comparisons used Kuperberg's bound and Klartag's covariance framework, while a projection–section calculation used the Rogers–Shephard inequality [39, 38, 58].

For a simple polytope, it expressed the volume product through the relative entropies of incident facet-slack coordinates against independent exponentials. It explored a reversible walk on the vertices and the Lawrence volume formula, then used face flags to assemble simplices [41]. A lower bound that retained only flags anchored at the same vertex failed on regular polygons with increasingly many sides. Returning to a common decomposition of the whole body, it found an omitted inverse vertex probability. With π_i proportional to the determinant associated with vertex i , the revised identity had the form

$$n!|K||K^\circ| = \sum_i \pi_i^{-1} \mathbb{E}_\sigma \prod_j \ell_{F_j}(y_{S_j}).$$

The corresponding Jensen estimate contained twice the entropy of π . The assistant then noticed that vertices could coalesce while their determinant weights stayed positive: vanishing flag slacks drove the logarithmic estimate downward. It retained an exact face recurrence with relative-entropy corrections and looked for a bound that would accommodate this degeneration.

The next approach differentiated the volume polynomial with respect to facet offsets, invoking the Hodge–Riemann framework for simple polytopes [64]. Under an additional nonnegativity condition on repeated edge coefficients, the assistant derived that every two-dimensional face was a triangle or parallelogram. It then propagated the local edge structure to describe an affine product of simplices and checked its factored volume product. It identified signed repeated coefficients as the obstacle to applying this argument generally.

Gale duality recast a slack fiber using the density of a projected Dirichlet distribution and a sum of determinants over feasible bases. The assistant considered complementary simplex projections, reverse Brascamp–Lieb estimates, centroid-section bounds, and Vaaler's section inequality [8, 27, 65]. It found that separately bounding the density and determinant sum lost their dependence on the same configuration. A wall-crossing calculation then led it to conclude that the weighted determinant sum was maximal at the weighted center. Deleting one coordinate and integrating all slices turned the induction into a reciprocal-section integral requiring a factor tending to e . Along a chosen line of centers, the assistant recovered that factor in the near-maximal-asymmetry regime; for smaller asymmetry it recorded a deficit and returned to covariance and stationarity conditions.

17 Stationary hazards and a translation instability

The assistant returned to its simultaneous boundary coupling at a hypothetical global Mahler minimizer. It sought deformations whose costs were affine on both primal and dual faces, but dense incidence patterns and projectively unique polytopes obstructed the proposed shadow motions. It revisited the shadow-system arguments of Meyer–Reisner and Chen–Li–Xi–Xu in this connection [47, 19]. Facet hazards of exponential cone measures then produced a reversible graph with a prescribed drift. Filtering its stationary process recovered a uniform Dirichlet law for a simplex; the assistant still needed an entropy estimate for general cones.

It next expressed slicing induction through integrals of reciprocal polar section masses. Primal and dual fiber endpoints shared a projected variable, allowing it to compare their Laplace transforms. When only one facet coefficient was positive, a width variable W satisfied $\mathbb{E}W = 1/p$ and $\mathbb{E}W^2 = 2/p$, forcing $p \geq 1/2$. It could not guarantee such directions. Testing positive-semidefinite cones, it found large inflation between the normal measure and the uniform section measure:

VERBATIM EXCERPT

| Thus direct criterion fails globally for PSD cones!

It therefore sought volume-product gains outside this central-density criterion, returning to Kuperberg's symmetric estimate, Reisner's zonoid argument, hyperplane symmetries, and anti-blocking induction [39, 55, 7, 59, 46]. Its projection and configuration comparisons used Rogers–Shephard projection–section bounds [58]. In recalling its graph-volume argument, it also invoked their difference-body bound through the inclusion $S \subset K - K$ [57]. Its covariance normalization followed the isotropic-constant approach [38]. Applying reverse Brascamp–Lieb to normalized ray frames gave a determinant lower bound whose missing factor it hoped to recover from the coupling [8].

The assistant then studied the stronger functional objective

$$e^{-d} d! |L| Z(L^\circ), \quad Z(Q) = \inf_z \int_Q e^{z \cdot y} dy.$$

Facet conditioning suggested dimension induction, with conditional hazards constraining a rescaling of the projected generators. The assistant tracked local simplex determinants and graph rates, but nonsmooth changes of triangulation continued to block the proposed second-variation argument.

Translating L preserved its volume and isolated Z . Writing $Q = L^\circ$, it used the projective change $y \mapsto y/(1 - tx \cdot y)$. At an interior stationary translation it obtained $z = 0$; minimizing over the tilt then gave

$$\left. \frac{d^2}{dt^2} \log Z((L - tx)^\circ) \right|_{t=0} = -(d+2)x^\top \Sigma_Y x.$$

VERBATIM EXCERPT

| Thus stationary translation is **strict local maximum**!

It first explored whether the normalized-moment log-concavity attributed to Berwald extended jointly to dilation and moment power, to control the drift between weights [12]. For a boundary chord, it rewrote the second derivative as a residual quadratic moment. Joint log-concavity of lifted integrals weighted by $(\pm x \cdot y - t)_+^2$ led it to the more general inequality $g'' \leq (g')^2$, where $g = \log Z$. It concluded that $1/Z$ was convex along translations and pursued strictness at stationary points through a reflected-gamma comparison. It then tried to extend this argument to shadow deformations and to compare that convexity with the changes of primal volume across triangulation boundaries.

18 Scalar Gaussian progress and the matrix obstruction

The assistant continued the Meyer–Reisner translation approach [47] by moving every nonzero vertex of a pyramid. It initially expected the resulting Hessian equality to restrict the cone, then recognized that the motion was already a linear transformation:

VERBATIM EXCERPT

Right translation of all vertices equivalent GL! Thus Hess equality automatically for pyramid all x.

Moving the distinguished origin onto the base instead gave a lower-dimensional functional product. The assistant calculated a square-root dimensional loss and found that a ball did not supply the compensating decrease it wanted. It also derived a polynomial comparison between the value at the original vertex and at the centroid, but its attempts to combine this with Kuperberg's symmetric bound retained an exponential loss [39].

The Gaussian route balanced primal and dual covariances by optimizing a metric. The assistant used Brascamp–Lieb and Bochner estimates for metric concavity, considered Ehrhard's inequality, and revisited Barthe's direct/reverse comparison [24, 8]. Optimizing the metric added positive terms involving third moments to the tilt Hessian; fixing a common metric allowed anisotropic escape even for orthants. Quadratic penalties restored bounded minimization but weakened the spectral estimate. Recalled entropy arguments also used Berwald's moment inequality and transport–entropy formulations of inverse Santaló inequalities [12, 32, 26].

The assistant then concentrated on a centered random variable W with density proportional to $e^{-x^2/2-\phi(x)}$, where ϕ is convex and W has variance s . It compared this law with two Gaussian tails joined at a kink, matching its first three moments. It assembled sign-change arguments, differential barriers for a half-line Gaussian parameter, and polynomial coefficient estimates into an argument for

$$\text{Var}(W^2) + \frac{1 + B_s^{-1}}{1 - s} (\mathbb{E} W^3)^2 \leq 2s, \quad B_s = m_s^2 - s,$$

where m_s is the mean of a half-line Gaussian law with variance s . It described this as a proof of the scalar estimate, then tried to extend it through successive one-dimensional Gaussian smoothings.

Earlier optimized slices introduced additional mixed Hessian terms, and fixed generic subspaces again admitted anisotropic escape. The assistant connected the remaining covariance problem with Klartag's conic formulation [38] and returned to section arguments, including Meyer's method [46]. A dimension reduction still lost a factor of order $\sqrt{d}$:

VERBATIM EXCERPT

Can't ignore .5log d per step.

It next rejected its proposed single-cone matrix moment bound: the radial component already exhausted the allowance, while an asymmetric tangential component could remain. The subsequent calculations sought cancellation between the primal and dual third-moment tensors, Gaussian boundary stationarity, and possible Jordan-algebra structure when the projective Hessian was degenerate.

19 Activation transport and polar proximal maps

The assistant returned to Gaussian regularization and the isotropic-constant approach associated with Klartag [38]. Rank-one elimination led back to a one-dimensional functional Mahler inequality, but exchanging a supremum and an integral introduced a loss from varying conditional maximizers. It revisited the transport–entropy formulation [32, 26] and tried averaging directions across independent copies. Its Gaussian approximation produced a criterion involving directional Fisher information times variance; steep approximations to exponential laws defeated that criterion.

VERBATIM EXCERPT

| For sharply truncated Exp, Fisher huge. CLT route fails.

Random perturbations of replicated tilts removed a first-order metric response, but the assistant found that the same perturbations lowered the objective at orthant equality and therefore required confinement or boundary control.

It next organized shifted cone projections by expected active-face dimension. A Legendre transformation of their cost profiles converted the desired estimate into an integral comparison with the orthant. The assistant calibrated a heat-flow trajectory to that reference, then found that its adjoint weighting measure was not even, whereas complementary terminal projections paired opposite shifts. A cosine seed density formally removed an asymmetry but introduced bounded-support requirements. It also obtained a Gaussian projection bound from a positive tight frame and asked how such a frame could be supplied in the dual cone D . Narrow circular cones obstructed that representation; the assistant argued that simultaneous positive tight frames on both polar sides would force orthant structure. Reflected Brownian motion suggested another construction through boundary local times, but singular distributions and dependence between stopping clocks prevented the desired entropy estimate.

Returning to moments, the assistant expanded the joint objective to fourth order at covariance equality. Cubic perturbations of a ball then led it to reject a universal covariance-product upper bound without additional shape conditions. It considered Caffarelli's contraction theorem [17] for controlling the third-moment tensor, and an algebra built from that tensor, but could not obtain positivity of the relevant multiplication operator. It also recalled local minimality at the simplex [37] while considering shape variations of the projection profiles.

The final branch used polar proximal maps, writing

$$y_\tau = \Pi_{\tau K^\circ} G, \quad x_\tau = G - y_\tau = \text{prox}_{\tau p_K}(G).$$

For a fixed Gaussian vector, the assistant related the dilation parameter to the gauge of the residual and interpreted the same residual as projection onto a dilation of K . Adaptive matching of the two gauge costs yielded an entropy expression for the distribution of logarithmic scales. It investigated whether changes of metric, determinant estimates, and strictness in gauge subadditivity could control broad scale distributions. It asked whether the resulting homogeneous gauge was convex.

20 Centered kernels and a sharpness gap

The assistant translated the Gaussian input to make the primal proximal image have mean zero. Writing P for its Jacobian, $B = \mathbb{E}P$, $t = \text{tr } B/n$, and $p = \mathbb{E}\langle x, y \rangle/n$, it derived $dt/ds = p$ even with the moving center. Gaussian chaos estimates gave

$$C = B - \text{Cov}(x), \quad \frac{1}{2}B(I - B) \leq C \leq B(I - B),$$

and it interpreted small p as proximity to complementary linear projections. It then corrected its treatment of convex envelopes: averaging slopes could increase the logarithmic slope integral needed for the entropy bound.

The assistant considered extending Borell's Gaussian noise-stability inequality through the semigroup approach of Mossel and Neeman [14, 48]. It questioned whether noncommuting Hessians admitted the scalar comparison, testing Clifford and Jordan examples and the restrictions imposed by symmetry of third derivatives. It also considered Gaussian comparison and regular ellipsoid positions [30, 52].

A cross-scale calculation instead led it to the kernels

$$m(t, u) = \min(t, u) - tu, \quad k(t, u) = \sqrt{p(t)p(u)} \exp\left(-\frac{1}{2} \left| \int_t^u \frac{dv}{p(v)} \right| \right).$$

Assuming $B_t = tI$, it used second-order Gaussian Poincaré estimates to obtain $m \leq 2k$ against nonnegative weights. It identified k with the Green kernel of $-d^2/dt^2 + V$, where $V = (1 - p'^2 + 2pp'')/(4p^2)$. For the orthant profile, $I(t) = \phi(\Phi^{-1}(t))$ satisfied $-I'' = 1/I$, giving equality in the associated Hardy inequality. The assistant then calculated the entropy variation and found that its weight was not proportional to I^2 :

VERBATIM EXCERPT

| Ratio varies ~.5 tails >.386 center. Not stationary $\Rightarrow$ Hardy alone insufficient even close to orth if V redistribution.

It pursued higher Gaussian chaoses, approximate nesting of Hessian projections, antipodal comparisons, and metric changes intended to remove the linear Gaussian component.

The assistant next tried to exploit the exponential law as the sum of two squared Gaussians. Integrating one-sided proximal paths and pairing opposite Gaussian inputs reproduced the orthant's quadratic map. For general maps, it encountered a change-of-variables obstruction:

VERBATIM EXCERPT

| General can have singularities and many preimages (non-gradient in r).

It investigated complex quadratic forms and self-dual cone order intervals. It revisited its use of the Rogers–Shephard difference-body bound [57]. In revisiting these routes, it recalled Kuperberg's linking construction, Klartag's covariance approach, reflection and zonoid inequalities, reverse Brascamp–Lieb, and Zhang's projection inequality [39, 38, 7, 55, 8, 67]; it also returned to contraction transport, Gaussian tube concavity, and polynomial-degree formulas [17, 24, 11]. Finally, it rewrote the calculation directly for cone projections with means M, N , obtaining $dt = (N \cdot dM - M \cdot dN)/d$, and explored paths with nonconstant mean directions to adjust the cross-kernel and entropy costs.

21 Entropy concavity, convexification, and fixed tests

The assistant rewrote the Gaussian projection slope entropy using $g = A'$, $h = -B'$, and $r = g/h$. The Wronskian identity made $Q(t, v) = g(\min(t, v))h(\max(t, v))$ doubly stochastic. A relative-entropy variational formula then expressed $J = \int \log(gh)$ as a concave functional of $\log r$, with an orthant tangent involving a weight that changed sign. It tried harmonic-mean estimates for the averaged projection Hessian, but the sign change prevented the proposed direct comparison. Convexifying the costs created another obstruction:

VERBATIM EXCERPT

| But total Q mass slack may offset? $J_{\text{env}} \geq J_{\text{raw}}$ usually, cannot assume.

The assistant next convexified the planar curve $B(A)$, mixing complementary projection maps at chord endpoints. It obtained a modified Wronskian density $0 < w \leq 1$, monotone virtual slopes, and a dual Neumann Green operator $-\partial(V^{-1})\partial + w$. Assuming a full Hardy inequality for V , it derived a comparison reducing the entropy expression to $w = 1$. It pursued metric balancing through truncated positive

eigenfunction tests and rank-dependent bounds on the costs, while retaining questions about continuity, singular measures, and endpoint limits. A positive-frame comparison using reverse Brascamp–Lieb gave a one-sided potential bound, whose entropy variation again changed sign [8].

Under the Hardy assumption, its orthant tangent estimate still left a positive gap. The assistant therefore studied near equality: an integrated Hessian close to a linear Gaussian matrix, nearly nested projections, and a contact kernel that separated into a product off the diagonal. Wick expansions and a quadratic test gave different coefficients for variance heterogeneity, which it hoped to turn into a quantitative contradiction. It considered local minimality at the simplex [37], but noted the dimension dependence of the required neighborhood. It also revisited scalar noise stability, Gaussian tube inequalities, and the loss in a symmetric-suspension comparison [14, 24, 39].

To make the perturbation more explicit, the assistant replaced the potential-dependent eigenfunction by the fixed Gaussian profile $I(t) = \phi(\Phi^{-1}(t))$. With $T = 1 - \mathcal{L}_G$, $\delta = I^2 V - 1$, $u = T^{-1}\delta$, and $\eta = zu - u'$, its balanced test gave the quadratic condition $2\mathbb{E}\delta + \mathbb{E}\eta^2 \leq 0$. Polynomial tests suggested a way to control anisotropy, but it still needed bounded tests and tail estimates. The assistant then noticed that this single fixed test did not provide the full Hardy spectral gap used in the convexification comparison. It explored hybrid tests combining the fixed profile with a positive eigenfunction, including how their mixed terms and balancing conditions would change.

22 Resolvent cancellation and mismatched Gaussian paths

The assistant decomposed an integrated Hessian as a linear Gaussian matrix plus higher Gaussian chaoses. It used eigenvalue-pair measures to compare matrix Dirichlet energy with a scalar Stein form, then sought to cancel first-order residuals through an auxiliary primitive whose derivative was a squared test derivative. It found that the proposed test class could fail to remain regular for induced measures with atoms, and returned to smooth Gaussian-reference tests. A direct spectral-layer calculation then canceled several nesting-error terms.

The assistant next considered multiplying a hypothetical counterexample by many orthant factors. It expected the nonorthant contribution to become a small perturbation, making squared displacement terms negligible to first order. Rewriting this calculation directly on the original cone produced proposed identities relating contact, nesting, and Gaussian-chaos errors. The assistant still needed a dimension-independent bound for the spectral trace error. Commuting Gaussian-mixture models exposed variance heterogeneity; noncommuting matrices added divided-difference errors. It tried logarithmically oscillating tests, complementary squared derivatives, and separate estimates for even and odd Gaussian degrees. It also found that some additional compatibility identities merely restated relations already available.

In these comparisons, the assistant reused the projective covariance restrictions associated with Klartag’s conic approach, revisited Borell’s scalar noise-stability comparison, and recalled why Kuperberg’s symmetric bound already exceeded the nonsymmetric asymptotic constant [38, 14, 39]. It returned to its earlier Gaussian projection and heat-calibration calculations rather than obtaining the remaining trace estimate.

A sign-changing entropy weight prompted a different path construction. The assistant tried splicing a primal-centered path to a dual-centered path; at the central junction, however, its Fenchel calculation produced the opposite boundary sign from the one it wanted:

VERBATIM EXCERPT

| Boundary ≥ 0 UNFAVORABLE!

For a straight aligned path it derived an expression involving expected Jacobian trace minus longitudinal

curvature, weighted by an odd function. It found that extreme horizontal stretching and compression could both make the expression unfavorable. Reversing the splice gave a favorable central boundary term, but the assistant still sought control of the negative Gaussian projection energies. It considered positive tight frames and reverse Brascamp–Lieb bounds as possible compensation for a poor dual profile [8].

Finally, the assistant separated the two centered paths altogether, prescribing primal and dual means independently and driving their projections by opposite Gaussian inputs. It noticed that their cross-cost bound now used the unperturbed orthant kernel:

VERBATIM EXCERPT

| This is huge: cross-cost kernel is *exactly orth reference*, not perturbed.

It began combining this contact bound with Hessian cross-products to seek the integrated rank comparison required by its entropy estimate.

23 Stitched kernels and a failed Gamma interpolation

The assistant joined a primal-centered Gaussian projection path below zero to a dual-centered path above zero. It found that ordered contact products retained the form $A(z)B(v)$ for $z \leq v$, even when the path jumped at the junction. Fixing A on one half and B on the other eliminated the quadratic perturbation of this kernel. Writing $J(z)$ for the sum of the two scalar barrier differences on the negative half-line, it expressed the primary deficit as

$$S = -2J(0) - 4 \int_{-\infty}^0 zJ(z) dz.$$

The entropy expression used a different weight. The assistant obtained a favorable junction term from the Fenchel gap, but still needed to control a trace defect associated with unequal Gaussian variances. Adding the decaying homogeneous solution $(A_0/\varphi)^2$ preserved the bulk differential identity; its even extension introduced a cusp and a single-layer term that the assistant could not bound by the existing deficit. It also explored staggered switches through increasing subspaces.

The assistant then revisited Gale densities, orthant sections and projections, and direct and reverse Brascamp–Lieb comparisons, including Barthe’s transport argument [16, 8]. Its recalled alternatives included Klartag’s covariance condition, Kuperberg’s symmetric bound, zonoid volume products, and shadow-system convexity [38, 39, 55, 47]. It also considered Vaaler-type section estimates, centroid-density estimates and log-concavity, while identifying losses in the resulting density bounds [65, 27, 54]. Separate direct and reverse inequalities bounded their integrals in opposite directions; the assistant sought a comparison of their deficits instead.

Centered Gamma potentials suggested such an interpolation. For normalized rays q_i , positive weights p_i summing to d , and dual cones C, D , it defined

$$P_D(x) = \prod_i (q_i \cdot x)^{p_i}, \quad P_C(y) = \sup_{y = \sum_i p_i q_i s_i} \prod_i s_i^{p_i},$$

with positive s_i . It divided the product of the corresponding weighted exponential integrals by $\Gamma(a+1)^{2d}$, calling the result $R(a)$, and asked whether it decreased to its Gaussian limit. Laplace expansions, Gamma Stein identities, and variations of the weights produced further conditions to check; the assistant also considered Ball’s cube-slicing bound as a possible density estimate [3]. Rare-tail comparisons for extra rays then suggested that $R(a)$ could fall below one for arbitrary balanced bases, including choices the

assistant had hoped to justify by determinant optimization. It therefore redirected the argument toward small positive powers and optimized weights.

Splitting a ray could produce a negative direct-integral variation of order $|t|^{1+ap_i}$; the assistant compared this with dual gains governed by masses outside adjacent facets. It still needed control of the optimizing weights and of existence when the number of rays increased.

VERBATIM EXCERPT

| If m unrestricted, existence may nonpoly.

24 Canonical barriers and affine-sphere volume targets

The assistant first paired Gamma potentials through a stochastic channel on normalized cone rays. Relative-entropy contraction ordered the potentials, but the two partition functions still used different Gibbs measures. Gale coordinates then expressed the volume product through feasible bases and projected slack volumes. The assistant explored centroid-section estimates, complementary projections, and random shifted constraints, drawing on Fradelizi's section bound and Rogers–Shephard comparisons; it found that independent thresholds could discard too much determinant mass for round bodies [27, 58]. Regular upper hulls gave an exact slope-density identity, which the assistant recognized as another form of its separable gradient map. It also revisited stationary covariance constraints from Klartag's approach and reverse Brascamp–Lieb estimates [38, 8].

The cubic term in the Gamma expansion suggested replacing a chosen logarithmic barrier by the canonical potential F , with

$$F(ty) = F(y) - d \log t, \quad \det D^2 F = e^{2F}.$$

Using the Cheng–Yau affine-sphere framework, the assistant sought the stronger comparison $\chi_C(x) \geq e^{F_{C^*}(x)}$, where χ_C is the cone Laplace integral [20]. For a cone over $K \subset \mathbb{R}^n$, with $d = n + 1$, it rewrote this as a maximum-height bound for the positive concave solution

$$\det(-D^2 u) = u^{-n-2}, \quad u|_{\partial K} = 0, \quad \|u\|_\infty^{n+1} \leq \frac{n!|K|}{(n+1)^{(n+1)/2}}.$$

The assistant checked simplex equality and ball slack, then tried to evolve the volumes of the level sets of u . Its proposed differential inequality required their normal speed to be a support function; examining simplex gradients overturned that requirement.

VERBATIM EXCERPT

| Thus p is *far* from support precisely near simplex, ODE fails drastically, can't bootstrap.

It next integrated the Hessian along a cone ray, $B(y, a) = \int_0^\infty D^2 F(y + ta) dt$, obtaining a chart that becomes coordinatewise logarithms on the orthant. Invoking Calabi's nonpositive-Ricci estimate, it derived determinant and trace comparisons, then sketched a cubic-tensor calculation using completeness and an Omori–Yau maximum principle [18, 66]. In normalized horizontal coordinates it obtained a convex function f satisfying

$$\det D^2 f \geq e^{-f}, \quad \Delta f + \frac{|\nabla f|^2}{d} \leq n.$$

The remaining target was a lower bound for the volume of $\nabla f(\mathbb{R}^n)$. The assistant related the two deficits through a radial differential identity and considered whether a nearly vanishing deficit forced simplex structure. Gaussian entropy trials retained an exponential loss. Marginalization trials used Prékopa and Brascamp–Lieb estimates; the assistant also returned to Bergman-kernel and Kuperberg comparisons while looking for a different integral argument [54, 16, 50, 39]. Finally, ellipsoidal fibers over the affine-sphere graph led to weighted Gamma integrals and a proposed monotonicity question.

25 Canonical comparisons tested beyond the orthant

The assistant pursued the orthant-calibrated inequality

$$\chi_\nu(a) = \int_C e^{-a \cdot y - (\nu-1)F(y)} dy \geq \Gamma(\nu)^d$$

for a normalized canonical potential. It compared angular integration by parts with the required trigamma variance, using Brascamp–Lieb estimates and considering Berwald moment normalization; its first gradient-moment test left a gap [16, 12]. A translated-barrier vector field reproduced the orthant identities, suggesting a pointwise divergence bound. The pointwise divergence bound remained unproved; the assistant turned to weighted or cumulative averages of its deficit.

A beta-convolution route paired complementary points and introduced an even potential whose orthant model was $\sum_i \log \cosh s_i$. For the paired averaged Hessian, the assistant used a determinant comparison it had derived assuming Calabi’s nonpositive-Ricci bound [18]. Lorentz-cone calculations reversed the proposed trace inequality, while Gaussian trial measures left a nonzero gap even for the orthant. The assistant considered Vaaler’s cube-section bound, revisited Kuperberg’s graph construction, and explored diffusion and integrated-Hessian coordinates as alternative ways to compare the integrals [65, 39].

Next it sought an exponential-to-Gaussian quantile map built from positive mixtures of translated Hessians. The local second variation had the desired sign under the same Calabi curvature bound. The global transport comparison remained unresolved, and the assistant considered radial corrections, determinant mixtures, and an Onofri-type energy comparison.

Laguerre tests led to a two-point Hessian target,

$$\mathrm{tr}(M(p)^{-1}M(q)) \leq p^T M(q) p.$$

The assistant derived a gamma comparison that would follow from this bound. Maximum-principle calculations gave signs the assistant could not use. Linearizing the orthant potential then produced mixed signs for some proposed deformation modes; it treated their realization by actual cones as a further question.

The assistant shifted toward averaged deficits and complementary pairs, while also revisiting shadow-system convexity, reverse Brascamp–Lieb transport, and Suita-type kernel estimates [47, 8, 50]. Gamma-ratio calculations suggested positive integral corrections despite the changing pointwise signs. For a symmetric pair it proposed summing the two trace deficits. It continued by asking how the paired inequality would enter the beta-convolution argument.

26 Toric residues and random supporting planes

The assistant examined a symmetrized barrier comparison and noticed that boundary terms prevented its proposed specialization to uniform density. It returned to Calabi’s nonpositive-Ricci bound in its trace comparison, but recognized that the Laplacian control did not supply the directional convexity it wanted [18]. It then represented a rational cone by height-one lattice rays, writing $N = \chi_C(v)$ and seeking $N\chi_D(u) \geq 1$ when $u \cdot v = d$. The multiplicity–log-canonical-threshold inequality suggested a toric analogue, but it found obstructions to finite smooth covers and to applying local multiplicity estimates on resolution charts [25].

A determinant polynomial led to zonotope and random-simplex comparisons. The assistant rejected a proposed reciprocal-volume bound after checking its dimensional scale for a ball. It next used polynomial degree and toric residues to formulate a sufficient norm–residue estimate [11]. One-dimensional examples exposed cancellation and conditioning problems. Replacing the signed trace by geometric

means of inverse Jacobians led to entropy criteria; a linear exponential-map criterion then failed for crosspolytopes. Its related comparisons revisited reverse Brascamp–Lieb, antiblocking, central-section, and reverse-isoperimetric bounds [8, 59, 65, 2].

The assistant assigned independent exponential heights to vertices and sampled a cell of the lower regular triangulation by its volume. With rays $r_i = \langle p_i, 1 \rangle$ and rates satisfying $\sum_i \alpha_i = d$ and $\sum_i \alpha_i p_i = 0$, it set $M = \sum_i \alpha_i r_i r_i^\top$. It calculated

$$\Pr(\text{sampled supporting plane is nonnegative on } P) = \frac{\det M}{N} \chi_D(u),$$

so its target became a lower bound of $\det M/N^2$ for this probability. Sparse rates returned it to the original polar integral. An event using the first Poisson arrivals yielded only a slab estimate; comparing that slab with a difference body and Kuperberg’s bound retained a loss [57, 39]. For random upper constraints it also obtained $\mathbb{E}|Q(b)| = \chi_D(u)$, but recognized that these sections passed through a corner, preventing a direct use of central cube-section bounds.

The assistant recalled Banaszczyk’s Euclidean transference bounds and Gaussian method [4]. Its lattice argument suggested another sufficient asymptotic constant, which it questioned through simultaneous packing and simplex examples. It examined Hilbert series, toric wall relations, and radial variations, while recalling the covariance and shadow-flow approaches [38, 19]. The crosspolytope obstructed its hope that radial stationarity would force a simplex:

VERBATIM EXCERPT

| If global min proof must compare all stationary including Hanner.

It also revisited volume localization, polynomial-capacity bounds, and Bergman-kernel comparisons [41, 34, 50]. Finally, it decomposed the random section’s normal cones by upper and lower active constraints. The lower-dimensional terms entered with the opposite sign from the one it needed for induction. Pairing primal and dual random boxes produced covariance determinants; its spherical calculation contradicted the proposed guaranteed existence of a mutually maximizing pair with only upper constraints active.

27 Random optimization and Gaussian replacement formulas

The assistant represented a random linear program on a dual cone by independent exponential constraint heights. Conditional on an optimizer stratum with k active upper constraints, it derived a gamma law for the optimum and a uniform law on the corresponding normalized face. Rescaling the heights made the survival event a Minkowski sum of a simplex section with the positive orthant. Its survival polynomial then had normalized mixed-volume coefficients; the highest coefficient measured the probability that the optimizer avoided the cone boundary. Complementary section–projection comparisons using Rogers–Shephard and moment comparisons using Berwald lost the constants the assistant needed [58, 12].

It next combined reverse Brascamp–Lieb estimates with a contact coupling having zero means and prescribed cross-covariance [8]. A test using two identically oriented simplices satisfied these moment conditions while omitting vertices of the actual polar. The omitted polar vertices showed why these moment conditions alone were insufficient. It examined an associated positive transition matrix, anti-blocking bodies, and the convex hull of contact pairs, comparing these constructions with Saint-Raymond’s volume inequality and Kuperberg’s linking-integral approach [59, 39]. Surface-normal covariances and averaged Hessians suggested another interpolation; projection estimates again left a dimension-dependent loss. Its return to Gaussian projections also recalled the covariance condition associated with Klartag’s isotropic-constant approach [38].

For these Gaussian spectral comparisons, d denotes a scalar profile independent of the cone dimension. With standard Gaussian density ϕ and distribution function p , write

$$X = p/\phi, \quad Y = (1 - p)/\phi, \quad f = -(1 - p) - \log p, \quad j = -p - \log(1 - p).$$

The profile is

$$d(x) = (1 + xX(x))^2 f(x) + (1 - xY(x))^2 j(x).$$

For a linear Gaussian symmetric matrix L , the remaining spectral term was

$$\Delta_\psi = \mathbb{E} \tau \psi(L) - \mathbb{E} \psi(G), \quad \psi'' = Nd,$$

with standard Gaussian scalar G , normalized trace τ , and Gaussian number operator $N = x\partial_x - \partial_x^2$. The assistant explored bounded diagnostic functions, changes of metric, and exact centering of the projection path. In a commuting model, exact centering appeared to force equal variances. It then noticed that small matrix errors could contribute coherently across coordinates, so this argument did not immediately give the desired quantitative bound.

Gaussian doubling suggested expressing the spectral discrepancy through two independent replacements. Averaging over variance splits removed the oscillations of a fixed split and gave an auxiliary profile

$$\varphi''_* = (N + 4)(d - 1/2).$$

In the commuting case, the assistant obtained quadratic Fourier expressions in mean characteristic-function discrepancies. In comparing noncommuting sums with spectral-overlap averages, the assistant invoked Golden–Thompson for real exponentials and hyperbolic cosines, then explored analogous trace comparisons for its profile [29, 63]. It also inferred a square-root trace comparison from Golden–Thompson through a Laplace integral and considered combining it with a lower Khintchine estimate [29, 63]. The assistant returned to the exact layer and contact identities, checked their algebra, and pursued a Duhamel expansion in which trace symmetry canceled a first-order commutator term. It still needed an estimate tying the remaining commutator terms to its nonnegative defects.

28 Commuting evidence and Hessian structure

The assistant pursued a dimension-independent bound for the Gaussian spectral discrepancy, writing $L = \sum_i G_i M_i$ and $S = \sum_i M_i^2$. It compared scalar and matrix Ornstein–Uhlenbeck operators through the defect

$$\tilde{d}_h = (1 + N_G)^{-1} (N_z h(L) - N_G h(L)).$$

It hoped that this defect, together with the Dirichlet gap J_h , would control variance heterogeneity and noncommutativity. Revisiting its own earlier calculations, it considered quadratic maps from pairs of Gaussians, tail constraints, and alternative metric balances. A rotating-projector example led it to reject a trace-norm reconstruction bound for arbitrary projection layers and to seek additional control from cone contact positivity.

Perturbing the cross-covariance of two Gaussian inputs, the assistant derived a quadratic-test inequality in which a residual square cancelled. It then encountered unbounded test functions, the restriction to positive covariance decrements, and difficulties controlling matrix errors. It proposed allocating diagnostics across dyadic variance scales, but observed that compressing away high-variance directions could reintroduce large derivative errors. In the commuting case, it considered a two-component profile involving hyperbolic functions.

The assistant next tried to change the balancing condition so that the remaining even discrepancy would disappear. Its Hermite calculation produced a positive adjoint compatibility condition, which it

took as an obstruction to a monotone replacement. A further proposal to offset the mean by a nonlinear scalar balance failed its own test at extreme scalings. It also derived a resolvent identity expressing a noncommutative comparison error as a product of commutators, then found that spectral cancellations obstructed the desired localization estimate. Revisiting Golden–Thompson, the assistant compared a square-root matrix trace with a spectral-overlap average, but cautioned against assuming the desired deterministic sign [29, 63]. Hermite triangularity suggested formulas quadratic in lower-chaos leakage; the assistant found that positive factorizations introduced polynomial tails incompatible with bounded diagnostics.

Returning to the full symmetry of the coefficient tensor, the assistant noticed that the comparison field in d_h was a Hessian. It therefore used the failure of $\nabla h(L)$ to be fully symmetric as an additional contribution to the Dirichlet defect.

It then investigated calibrating S , diagonalizing an independent Gaussian reference matrix, and factoring a classical two-variable comparison kernel. It continued to identify anisotropic variance, transfer to actual projection layers, and global estimates as obstacles to its proposed bound.

29 Parity estimates and the limits of path averaging

The assistant tried angular averaging to factor the Gaussian trace discrepancy into squares of mean discrepancies. Hermite–Laguerre expansions suggested positive moment kernels, but its Taylor expansion gave signs incompatible with complete monotonicity of the required scalar profile. It therefore considered signed representations and approximate profiles. Recalling its reverse Brascamp–Lieb and affine-sphere routes, it again found missing comparisons between the primal and dual quantities [8, 18].

It next tried bounded non-Gaussian inputs so that linear diagnostics would remain bounded. For a cosine density, it obtained a constant product of the two scalar hazards. It then noticed that the centered primal mean reached a finite endpoint while the dual hazard diverged, potentially making the dual entropy cost infinite. Logistic inputs gave complementary derivative matrices, but the cost comparison remained unresolved in its calculation.

Returning to Gaussian inputs, it split the higher-chaos remainder into even and odd parts. The even part began at degree two and the odd part at degree three; the assistant used these different degrees to assign separate quadratic estimates. It retained more of the companion diagnostic’s square instead of replacing everything by a constant bound. Variance-dependent Stein equations and threshold decompositions then targeted the remaining matrix errors. One threshold Cauchy–Schwarz estimate developed a logarithmically divergent tail.

Hessian symmetry supplied another term: the assistant compared divided differences along the three edges determined by a symmetric third-order tensor. It recognized additional local control from the part of a diagnostic that was not itself a Hessian:

VERBATIM EXCERPT

| Thus Codazzi doubles J locally!

In comparing matrix and commuting Gaussian laws, it considered Golden–Thompson and Lieb trace inequalities, then explored an averaged commutator-moment estimate near identity covariance [29, 63, 42]. Star-shaped matrix examples obstructed its proposed extension to heterogeneous covariance; its attempted use of Lieb’s concavity for skew information likewise left a scaling problem.

The assistant finally revisited paths that switched from primal centering to dual centering at a cutoff. It derived the cutoff boundary terms and asked whether averaging the cutoff could remove the troublesome profile. Matching the coefficients of both barriers forced their averaged profile back to the original one:

VERBATIM EXCERPT

| So smooth profiles no help;

It then expressed the Fenchel gap as a squared mismatch of centered projections plus a nonnegative contact term. From an identically zero gap it recovered the Gaussian rank profile. For small gaps, however, it noticed that scalar barrier perturbations could change the entropy expression at first order while changing the gap only quadratically. It turned to the vector mean constraints and independently centered primal and dual maps for additional control.

30 Variance rigidity, large spectral rows, and smoothing

The assistant compared independently centered primal and dual projection paths, then tried a deterministic shift with a common mean residual. Cancelling its integral in a commuting model forced unit spectral variances, but it still needed a calibration realizing that cancellation. Anisotropically scaled orthants defeated the proposed entropy tangent. A contact identity relating mean rank to covariance suggested another route: exact reflection parity would force the Gaussian profile. Its quantitative version encountered a positive adjoint compatibility condition, while scale-dependent Fourier tests recreated the unbounded triangular-kernel obstruction.

The assistant next asked whether high-variance rows of a Gaussian matrix could retain a definite aggregate fraction of their variance in the diagonal of $\mathbb{E}|L|$. Star-shaped examples ruled out the individual-row estimate. It considered noncommutative Khintchine duality [45], then grouped variances by dyadic scale and charged interactions with larger rows through their rank.

Positive-part spectral tests offered a second approach, using the Powers–Størmer inequality and Douglas factorization [53, 23]. The assistant estimated the constants needed by its entropy argument and found that a small positive universal bound would not suffice.

It returned to Gaussian regularization of cone Laplace integrals. Optimizing primal and dual metrics matched posterior covariances, but the resulting Hessian contained third-moment corrections. Reexamining its earlier scalar truncated-Gaussian argument, it distinguished that calculation from the multidimensional bound it still sought. The recalled approaches used Ehrhard’s inequality and Brascamp–Lieb estimates [24, 16], Caffarelli contraction and Borell noise stability [17, 14], and Berwald’s moment comparison [12]; the assistant continued to encounter tensor or conditional-centering obstructions.

Its review also returned to Klartag’s covariance framework [38], canonical Cheng–Yau barriers and Calabi curvature estimates [20, 18], and reverse Brascamp–Lieb bounds [8]. Kuperberg’s symmetric bound and Saint-Raymond’s inequality remained possible inputs to lifts [39, 59], but the assistant noted that fixed exponential losses prevented its proposed transfer from being sharp at the simplex.

Moreau decomposition expressed the product integral through translated primal–dual intersections. The assistant checked that replacing each intersection by the fixed cone intersection was exact for the orthant, then found that this replacement could discard needed slack for other models. A proposed covariance monotonicity met a more explicit obstruction. For

$$C = \{(z_1, z_2, x) : z_1, z_2 \geq 0, 0 \leq x \leq z_1 + z_2\}, \quad \chi_C(a, a, \lambda) = \frac{2a + \lambda}{a^2(a + \lambda)^2},$$

it differentiated twice in a and then in λ . The variance of $z_1 + z_2$ increased when λ was sufficiently large, and it rejected the universal curvature principle:

VERBATIM EXCERPT

| Thus T_v not PSD globally.

The assistant then proposed realizing smoothing by higher-dimensional cone lifts, hoping to turn covariance improvement into amplification of a hypothetical deficit.

31 Canonical entropy and the failure of scalar induction

The assistant first reconsidered perturbations of product cones, then compared maximal entropy on a cone with its canonical potential F , normalized by $\det D^2F = e^{2F}$. Its intended sufficient inequality was $H_0(y) + F(y) \geq d$. It relied on Cheng–Yau completeness and a Calabi curvature calculation, and used Brascamp–Lieb variance estimates to organize Gaussian regularization at fixed mean into a positive trace term, a negative square, and variance deficits [20, 18, 16]. Completing the square still left aggregate control of third moments unresolved. Covariance caps made the covariance isotropic when every direction was active; inactive directions and mixed precision terms remained obstacles.

It then calibrated a spectral penalty against the half-line entropy profile. Testing off-diagonal precision matrices on an orthant reversed the desired comparison:

VERBATIM EXCERPT

| Thus J negative near non-diagonal orthant! Not valid for all Q .

The assistant next pursued one-dimensional slices. Retaining the variation of the maximizing eigenvector in a Bochner calculation led it to announce a two-functional bound. It used this to restrict reciprocal endpoint distances and attempted to compare interval entropy with half-line entropy, leaving an elementary hyperbolic inequality and a damping argument to prove. Partial Legendre elimination then exposed a failure of dimensional induction: an oblique first slice of an orthant gives a strict surplus, although the total orthant entropy is exact, so the remaining factor must fall below the proposed inductive target.

VERBATIM EXCERPT

| So iterative scalar $\geq e$ false.

Further covariance-cap calculations controlled radial third moments but left transverse contributions unresolved. The assistant explored approximate simultaneous diagonalization of the canonical cubic tensor, including a reformulation in terms of completely positive maps, to relate these contributions to curvature. It also revisited reverse Brascamp–Lieb estimates, hyperplane-symmetry arguments, and Saint-Raymond’s antiblocking inequality from its earlier calculations, without finding the missing comparison [8, 7, 59]. A gradient-coordinate approach led it back to moment-measure equations [21]. Its proposed containment by a spectrahedron defined by the cubic tensor failed on centrally symmetric nonspherical sections.

The final sustained pivot used the symmetric order interval $K = \{z : a \pm z \in C\}$ and a beta-integral comparison, with the orthant as the equality model. The assistant considered whether Kuperberg’s symmetric volume-product bound could handle sufficiently round cases [39]. Symmetry removed odd posterior moments, but determinant control alone admitted radial constructions below the intended constant. Splitting the averaged chord Hessian into two halves then gave the assistant a trace bound: after normalizing $D^2F(a) = I$, it obtained $\text{tr } P \leq d$ for the inverse averaged Hessian P . It ended by trying to combine this trace bound with determinant slack and finite-parameter beta estimates.

32 Beta convolution and failed positivity certificates

The assistant continued the canonical-potential route, invoking Cheng–Yau completeness, Calabi’s Ricci estimate and an Omori–Yau maximum principle [20, 18, 66]. With $F(a) = 0$ and $F''(a) = I$, it sought

$$I_\nu(a) = \int_{a \pm z \in C} e^{-2(\nu-1)U(z)} dz \geq B(1/2, \nu)^d, \quad U(z) = \frac{1}{2}(F(a+z) + F(a-z)).$$

It calculated that the corresponding convolution inequality would give $R_\nu^2 \geq R_{2\nu}$; a saddle-point limit would then give the desired cone bound. In its gradient coordinates, an inverse averaged Hessian P produced the correction $\theta = \log \det P + 2U$. Splitting the chord into halves expressed this correction through three nonnegative terms.

It hoped small slack would permit approximation by commuting matrices and the orthant’s product density, but still needed quantitative control. Optimized Gaussian trial laws did not reach the orthant constant at $\nu = 1$. A proposed pointwise bound demanded a fourth-derivative inequality the assistant could not justify. Vinberg-cone boundary asymptotics also led it to reject a Gaussian-quantile certificate for some exponents. Revisiting reverse Brascamp–Lieb and Saint-Raymond arguments did not supply its missing comparison [8, 59]; ordinary Brascamp–Lieb covariance control gave a weaker convolution-Hessian estimate even on the orthant [16].

Integration by parts next yielded an ordinary differential equation with an extra term whose sign would control $\mathbb{E}(2U)$. Formal perturbations of the orthant produced negative first variations. Integrating against a positive Green weight still left adverse boundary examples. It changed the determinant weight:

VERBATIM EXCERPT

| w=0.5 may fix both:

The half-weight represented Riemannian volume for the averaged Hessian. Subsequent spectral decompositions isolated positive logarithmic terms, while their expected values remained to be controlled. The assistant also withdrew a derivative estimate after testing the dilation direction:

VERBATIM EXCERPT

| Oops B_gamma HS bound impossible unless θ gradient zero:

It then used stationarity in the center to derive endpoint moment constraints, revisiting the covariance viewpoint associated with Klartag [38]. A Poisson equation represented the convolution Hessian through transport energy. Determinant concavity led back to an entropy expression with the opposite variational direction from the one needed.

At the uniform-density endpoint, the assistant’s gradient-covering argument likewise gave an upper bound. Kuperberg’s estimate left the same constant gap in its difference-body calculation [39]. It finally turned to varying the cone itself and wrote the transformed Monge–Ampère equation for shadow deformations, asking which comparison direction the fiber interpolation would provide.

33 Amplification and almost-pyramidal reduction

The assistant first tried a partial Legendre transform of a shadow deformation, but the resulting determinant comparison did not give the interpolation inequality it needed. It then sought to amplify a hypothetical Mahler deficit through products and canonical-potential truncations. An explicit barrier

gave a determinant estimate with a polynomial loss; the assistant noticed that the induced concentration bound became easier to satisfy when the characteristic volume was small. Dual truncation, viewed as primal expansion, led it back to the same direction of comparison. A collective concave constraint on a product cone gave a proposed multiplier involving a Chernoff bound and a weighted Minkowski volume. Radial integration expressed that multiplier through polar Minkowski sums, and the assistant recognized the original Mahler minimization inside the proposed amplification.

It next reconsidered random simplex determinants and stationary boundary couplings. Independent sampling lost the probability of selecting distinct simplex vertices, while squared determinants brought it back to covariance estimates. Its comparisons recalled Klartag's isotropic-constant framework, reverse Brascamp–Lieb and reverse-isoperimetric bounds, Kuperberg's symmetric estimate, and centroid-section estimates [38, 8, 2, 39, 27]. It also revisited shadow systems and the three-dimensional argument [47, 19]. While seeking a deformation beyond the available stationarity identities, it recalled curvature restrictions on local minimizers [56].

For a body close to a pyramid, the assistant chose opposite extremal points with asymmetry parameter $s \leq n$. A perspective change of the section height produced sublevel bodies $L_x = \{f \leq x\}$ and $M_y = \{f^* \leq y\}$ in dimension $n - 1$. It derived weighted formulas for both volumes and isolated a prefactor minimized at $s = n$. Applying the lower-dimensional Mahler benchmark left a product of expectations: primal mass sampled small heights, whereas the dual integral sampled much larger ones. Mixtures of directional thresholds defeated its first scalar comparison, so it sought compensation from the geometry of the level bodies and their centers.

With symmetrically rescaled heights g, h and $r = n/s - 1$, the assistant obtained an envelope estimate that avoided its earlier tail-divergence objection. It then revisited the scaling Hessian condition in the exponential limit and derived

$$\text{Var}(g) + \text{Var}(h) \geq 2r.$$

VERBATIM EXCERPT

| This kills mixed small-threshold model (both variances $O(r^2!)$);

It compared this constraint with information-content variance bounds [28], but rare large heights and the heights of contacting boundary points still escaped the control it sought. It pursued size-weighted contact couplings and clipping variations without obtaining the proposed near-pyramid induction. Finally, it turned to the nonnegative kernel $T(p, q) = 1 - p \cdot q$: its spectrum suggested a comparison between nearly Euclidean components and components resembling joins, while reverse Brascamp–Lieb still left a covariance determinant to recover.

34 Random heights and unused entropy slack

The assistant returned to contact geometry and rejected a fixed-sign rule for products of primal and dual simplex determinants: a four-by-four nonnegative slack matrix with zero diagonal could have the opposite determinant sign. It found that taking absolute values discarded cancellations needed at the simplex. Revisiting direct and reverse Brascamp–Lieb transport, it sought extra determinant gain from nonorthogonal frames, but found neither its Gaussian construction nor its inball estimate controlled the intermediate covariance regime [16, 8]. It also reconsidered the shadow-flow obstruction, Kuperberg's linking construction, and Rogers–Shephard comparisons [19, 39, 57].

Triangulating a cone led it to an entropy formulation. A basis had probability proportional to its absolute determinant, and its inclusion probabilities supplied rates for independent exponential heights. Integrating the height residuals related the probability of a lower supporting plane being nonnegative to the dual Laplace integral. For the resulting random linear program, it wrote:

VERBATIM EXCERPT

| Conditional on "good", density $\sim \exp(-D(u))$ independent of x !

The assistant still needed a probability estimate for the weighted triangulation, rather than a sum over all bases. It connected the construction to a zonotope and its normal cones, hoping to use the zonoid volume-product inequality, but suspected that its proposed local comparison merely restated the original difficulty [55].

It next sought a matrix extension of Borell's Gaussian noise-stability comparison [14]. Projection Hessians and their rank-one jumps across polyhedral walls offered additional structure. Scalar rearrangement gave directional mean bounds, but the assistant found that a positive decomposition of the identity using directions from both cones would already impose simplicial structure. It also recalled local simplex minimality, while noting the dimension dependence of that approach [37].

The assistant then audited its earlier Hardy–entropy argument: convexification of the cost curve, metric balancing, Gaussian Poincaré estimates, endpoint integrability, and smoothing of the transport maps. A nonzero entropy correction remained.

Returning to Klartag's projective covariance condition [38], it looked for unused relative entropy between its transport laws and tilted cone laws. Unequal coordinate rates might create a useful mean discrepancy. Splitting exponentials into independent gamma variables preserved the orthant distribution, but conditioning on their proportions produced Jacobians whose positivity and injectivity it could not control.

Finally, cumulative positive matrices gave it a proposed logarithmic-determinant comparison without simultaneous diagonalization. Its reaction was:

VERBATIM EXCERPT

| exact rearrangement! Orth equality when support pure threshold simultaneously; no dephasing. Nice.

It developed compact metric perturbations and positive spectral tail jumps, using conditional Gaussian averaging to estimate their entropy gain. Its remaining tasks included controlling finite perturbations, intermediate variances, tail constants, and the equality case.

35 Anisotropic inputs and quadratic layer control

The assistant returned to its earlier scalar Hardy and matrix-diagnostic calculations, seeking the entropy gain that those estimates had left missing. It examined constant and varying projection metrics, with the mean of each projected Gaussian still constrained to a prescribed ray. A constant metric gave a matrix diagnostic equal to the identity in the orthant model. The assistant sought a uniform bound for its response, especially on small subspaces with large covariance eigenvalues.

For a varying metric, it found that an odd coefficient would cancel the reflected covariance term of the ideal spectral-threshold family, including noncommuting thresholds. Transferring that cancellation to the actual projections led to inverse spectral-distance weights:

VERBATIM EXCERPT

| Log divergence near $z=\lambda_i$.

It tried subtracting threshold values, integrating matrix primitives at column-dependent cutoffs, and using nonnesting and commutator estimates. The assistant could not obtain the needed uniform estimate from these reformulations.

It then reconsidered polynomial diagnostics, alternative input distributions, and its earlier doubling argument for the Gaussian matrix series. Polynomial derivatives introduced weighted errors beyond the available deficit; smoothing changed the orthant equality calculation. The assistant also revisited cone-contact inequalities at individual pairs of thresholds, hoping their signs would distinguish mixtures of different variances. It repeatedly encountered transfer errors proportional to the square root of a deficit.

A change to the Gaussian input covariance produced a new first-variation calculation. For covariance $I + E$, it expressed the variation using only the mean projection Hessian H_z , with $j = A/\Phi$:

$$\tau E \left[\frac{1}{2} I - \int j(z) j'(z) (I - H_z) dz \right].$$

The assistant regarded this as a simpler moment diagnostic, but its attempts to control a finite perturbation encountered concentrated responses and tail terms. Matching the projection metric to the input transformation left diagonal orthant projections unchanged, which prompted it to seek estimates that retained this exact cancellation.

Rather than continue comparing two changing projection paths, the assistant rebuilt the complementary identities at an anisotropic covariance $\Sigma = I + T$. It separated the weighted trace defining the mean-rank measure from the ordinary trace used in the spectral-layer estimates. When it allowed $\tau \Sigma \neq 1$, it noticed that its formulas needed additional centering constants and corrected them. Positive covariance directions could be tested through cone contact; negative directions still required an upper estimate for a quadratic layer error.

For that error, the assistant subtracted a positive kernel within each side of a spectral threshold. It found that the remaining kernel decayed in the tails:

VERBATIM EXCERPT

| Thus no divergence for **quadratic** unlike saturated.

It then proposed controlling the remainder by the square root of the misalignment deficit times a commutator norm, and comparing that cost with the remaining entropy estimates. It continued working out this bound and its constants.

36 Riccati cancellation and explicit Gaussian profiles

The assistant continued the adaptive-covariance approach, writing the integrated projection statistic as $A = L + R$ and the covariance as $I + T$. It checked that the trace identities survived insertion of the covariance, but found that the weighted Jensen defect could have either sign. An even companion and bounded oscillating diagnostics were meant to control endpoint errors without consuming the available layer estimates.

A negative scalar Stein profile led the assistant to choose T through a matrix Riccati equation,

$$\lambda T^2 + \text{sym}(TC) = -D.$$

It derived a cancellation of the leading variance discrepancy and sought to use the resulting quadratic penalties to absorb the remaining terms. Moving the main quadratic layer calculation from L to the full statistic A let it charge an error to nonnesting instead of using only the smaller projection-mismatch allowance. It also retained the weighted residual term long enough to cancel part of the covariance correction.

The assistant next examined the scalar identities underlying the conic formulation. With the Gaussian ratios X, Y and logarithmic functions f, j defined above, the companion profiles are

$$g = \frac{1}{2}(1 - X^2 f - Y^2 j), \quad v = \log(XY).$$

For the Gaussian number operator N , it recovered $(N + 2)g = d$ together with

$$2g' = 2zd - d' - v'.$$

The assistant used this identity to express the resolvent profiles directly through Gaussian densities, distribution functions, and logarithms. In particular, it obtained $r_0 = -d - g''$, expressing the Riccati profile in explicit functions.

Its equality discussion depended on strict residual estimates: it reasoned that equality would force $T = 0$, commuting coefficient matrices, and vanishing projection mismatch. A coordinatewise cone projection would then identify an orthant and hence a simplex. It also revisited smoothing and boundary terms for arbitrary cones.

Next, a pointwise weighted Cauchy estimate combined the same-side and opposite-side layer integrals into a coefficient involving $\log 2 + 1/4$. The uniform scalar bounds remained to be established.

It explored small-interval derivative estimates, hyperbolic coordinates for the rotating diagnostics, and tail bounds from Mills-ratio integrals. The difficulty was to control intermediate intervals and cancellation in the tails. The assistant then sought analytic estimates for those remaining terms.

37 Scalar estimates and renewed algebraic scrutiny

The assistant pursued uniform Gaussian-profile inequalities in the coordinate $x = 3.6 \sinh t$. It used the profiles $K = d - 2g$ and $C = -d + 6.5g''$, also written K, C , and combined their Mills-ratio formulas with complex-strip estimates and analytic interpolation bounds. The proposed division separated short endpoint intervals, a compact region, and tails. Peano kernels improved the short-interval estimates; for the compact region, the assistant retained correlations between profiles instead of adding their separate maxima. It also corrected its interpolation plan to account for the fact that the two coordinate curvatures could attain their maxima at different points.

It then rederived the cone-projection and entropy identities before continuing the scalar estimates. In this audit it identified the positive covariance-contact term as the expected inner product of the appropriately ordered primal and dual projections, correcting a description in its own earlier reasoning. It checked the matrix Stein calculation's endpoint ordering: the outer eigenvalue supplies the endpoint value of C , while the shared eigenvalue supplies the endpoint of the squared probe difference. This ordering determined the sign in the asymmetric endpoint term. It also revisited the Riccati equation, commutator estimates, and the proposed deduction of orthant rigidity from equality.

For the scalar estimates, the assistant studied zero-free complex strips, Gaussian-tail bounds, and interpolation errors. It retained correlated profile estimates, which were needed to control the endpoint terms uniformly.

The assistant assembled the scalar estimates and added tail bounds to the argument.

38 A conditional proof chain and its remaining obligations

The assistant developed the segment estimate linking averages of the scalar profiles K, C and a rotating diagnostic $u = (\pi, q)$ to the variance of u . Here

$$b = .602, \quad c = .365, \quad k = \sqrt{b - c}, \quad r = .22, \quad q(x) = k - \sqrt{b - r^2 - d(x)}.$$

In the coordinate $x = 3.6 \sinh t$, the circular component is $\pi(t) = r(\cos(4.6t), \sin(4.6t))$. It separated short intervals, analytic tails, and a compact region, seeking uniform control of second derivatives and endpoint errors.

The assistant reconstructed the cone reduction and the Gaussian argument. For the cone over $K - z$, in dimension $m = n + 1$, it calculated

$$\chi_C(V)\chi_D(U) = \frac{(n!)^2|K|(K - z)^\circ|}{m^m}.$$

It sought a lower bound of one by integrating biased complementary projections against Gaussian hazard weights. It replaced direct differentiation of the full integrals by truncation followed by smoothing, then worked through the log-Jacobian surplus and the entropy costs. Its proposed surplus contributed $N/8$, where N integrated the nonnegative kernel measuring failure of the projection derivatives to be nested.

The subsequent reconstruction kept the covariance $\Sigma = I + T$, the change of metric, and the projection bias separate. The assistant used a simultaneous fixed-point argument to seek zero mean for the integrated derivative field and a matrix Riccati equation for T . It rederived the covariance, contact, and layer identities, checked the orientation of matrix indices, and split the residual into even and odd Gaussian degrees. Geometric subtraction on the two sides of a spectral threshold produced the constant $2\log 2 - 1/3$; another integral gave $\log 2 + 1/4$. These estimates were intended to pay for commutators and leave the segment variance available for the final comparison.

The assistant assembled a draft in which the Riccati identity supplied a quadratic penalty of at least $1.94 \tau T^2$, against a stated error allowance of $1.8 \tau T^2$. It argued that equality would force the coefficient matrices of the linear Gaussian field to commute and the projection mismatch to vanish, making the cone projection coordinatewise and its image an orthant. It then calculated the simplex volume product directly.

39 Recovering variance in the scalar estimates

For transformed endpoints $m - h, m + h$, with $x = 3.6 \sinh t$, the assistant used physical averaging with density proportional to $\cosh t$. Its circle $\pi(t) = .22(\cos(4.6t), \sin(4.6t))$ provided variance against which it charged the chord error of K , endpoint differences, and squared discrepancies involving C, q . Separate estimates lost correlations between derivative and curvature profiles. It tried joint rearrangement, Hölder bounds, and piecewise secant majorants for squared averages.

The assistant then returned to variance omitted from its sufficient inequality:

VERBATIM EXCERPT

| Maybe improve variance term by using discarded q variance.

It set $q_d = (q_+ - q_-)/2$, $e_q = \bar{q} - (q_+ + q_-)/2$, and $V_q = \overline{q^2} - \bar{q}^2$, obtaining

$$b_a = V_q + e_q^2 + q_d^2, \quad b_d = -2q_d e_q.$$

It observed that the endpoint penalty left a subtraction of V_q , while expanding the residual squares consumed only part of that subtraction under its proposed bounds. From a Lipschitz bound on q and a lower bound on the averaging density, it derived a cubic lower estimate for variance in terms of endpoint displacement. It also examined the extra circular variance proportional to $B_h^2 \operatorname{sech}^2 m$. These gains led it back toward simpler separate profile estimates.

The assistant next sought concentration bounds of the form $\int_E A \leq 2H(\alpha - \beta H)$ for $|E| = 2H$. It reduced them to excess integrals $\int (A - z)_+$, and used layer-cake integration to bound weighted averages. For larger intervals and intervals crossing zero, it expressed averages through weighted primitives and compared the endpoint correction with a one-variable function. It continued choosing thresholds and estimating chord errors, approximate monotonicity tolerances, and the remaining corrections needed to establish uniform bounds.

40 Bernstein positivity and the conditional cone reduction

The assistant sought scalar bounds through concentration estimates, primitive bounds, and polynomial inequalities. For middle widths $.16 \leq h \leq .54$, it retained the variance of q , obtaining

$$\frac{V_q}{h^2} \geq \frac{(h \coth h - h)z^3}{3 \cdot .193}, \quad z = \frac{|q_+ - q_-|}{2h}.$$

It used this cubic gain to absorb endpoint penalties, then multiplied the remaining deficits by $(\sinh h/h)^2$ to remove hyperbolic denominators. Taylor bounds reduced the remaining estimates to positivity of polynomials of degrees 30 and 32. It handled larger widths through concentration, monotonicity, and sign-dependent angular bounds. The required positivity of the resulting polynomials remained part of the scalar estimate.

It then reread its geometric argument from the beginning. With $m = n + 1$, the cone C over a translated body and its positive dual D reduced the proposed bound to

$$\chi_C(V)\chi_D(U) \geq 1, \quad U \in \text{int } C, \quad V \in \text{int } D, \quad U \cdot V = m.$$

The assistant checked biased Gaussian projections realizing prescribed means, their tail estimates, and a Brouwer argument choosing coordinates and a covariance perturbation. It corrected wording that appeared to give pr_1 the same total mass as $h'(z) dz$: the former was asserted to be integrable with all moments, while the latter had mass s_0 .

The audit followed the entropy estimate through truncated, smoothed projection maps, positive Jacobians, and Gaussian covariance identities. The assistant rederived the decomposition $A = L + R$, separated the even and odd parts of R , and checked how weighted Cauchy–Schwarz estimates and a matrix Riccati relation paid for the terms involving matrices that did not commute. The argument remained conditional on the scalar inequalities; the assistant continued checking signs, constants, and equality deductions.

For equality, the assistant reasoned that diagonal support of the endpoint measure would force the coefficients of L to commute, while vanishing layer error would make the projection derivative diagonal. It explicitly identified the next step:

VERBATIM EXCERPT

| Need argument all $j \neq i$ simultaneously yields dependence only i , continuity extends full slices.

It used mollification and vanishing cross derivatives to obtain coordinatewise dependence, then a product cone, an orthant, and a simplex section. Finally, it returned to complex-strip interpolation, profile derivatives, and the tail expansion.

41 Following the error estimates from tails to middle widths

The assistant continued checking the scalar estimates for its proposed projection-field argument. With $\xi = x^{-2}$ and $D = x\partial_x$, it differentiated Laplace integrals through order four and used binomial convolution to track products of derivative bounds. It checked the resulting estimates for K , d , and C , including the square root defining q . For the positive-part integral beyond 64, it used $K(x) \leq x^{-2} \log x$.

Under $x = 3.6 \sinh t$, uniform physical-length averages acquire a $\cosh t$ weight. The assistant integrated the circular component explicitly, obtaining a lower bound f_0^* for its variance, and derived a positive chord kernel for the averaging error. For half-width $h \leq .16$, it bounded the joint curvature and endpoint cost by $.54u_0$, where $u_0 = r^2 w^2 h^2 / 3$; its remaining allowance was approximately $.555109u_0$. It

then checked three tail configurations, using positive-part integrals, narrow profile ranges, and convexity to leave a bounded middle region.

For that region, the assistant used threshold integrals to bound concentration and integrated piecewise linear profiles to estimate weighted primitives. Almost-monotonicity estimates controlled endpoint discrepancies. Retaining the scalar variance V_q , rather than dropping it, gave

$$V_q/h^2 \geq \frac{v_h z^3}{3 \cdot .193}, \quad z = \frac{|q_+ - q_-|}{2h}, \quad v_h = h \coth h - h.$$

It reduced the excess cost to a cubic maximization and reported $R/h^2 \leq .027$. Hyperbolic and trigonometric expansions then produced deficit polynomials of degrees 32 and 30; it reduced their positivity on $[.16, .54]$ to Bernstein-coefficient inequalities. For larger widths it separated signs of the angular term and checked the rounded error bounds individually.

42 Checking transport, noncommutation, and equality

The assistant continued checking the scalar estimates and the cone argument. It bounded Gaussian tails, differentiated the profiles d, g, K, C, q , and examined the positivity needed before taking the square root in q . Concentration estimates and polynomial bounds were used to organize the remaining scalar inequalities.

It then rederived the cone normalization: with $m = n + 1$, positive dual cones and $U \cdot V = m$, the target became $\chi_C(V)\chi_D(U) \geq 1$. For the biased Gaussian projection field it checked coercivity, positivity of the smoothed Hessian, tail convergence, and continuity under changes of coordinates. The covariance choice $\Sigma = I + T$ used a matrix square root and a fixed-point argument; the assistant explicitly expanded the quadratic relation without assuming that its matrices commute.

Truncation and Gaussian smoothing gave the maps whose Jacobians entered its entropy calculation. It tracked the determinant factors, change of variables, conditional averaging, and integration-by-parts terms. The orthant served as a check: its projection layers became spectral thresholds, the residual vanished, and the proposed entropy identities matched the reference case.

The assistant also rederived the Gaussian covariance identities through Hermite expansions, introduced positive measures on layers and spectral pairs, and checked the commutator estimates used to absorb noncommuting errors. Its equality argument made the coefficient matrices commute, then made the projection derivative diagonal in a fixed basis. Vanishing weak cross derivatives led it to a coordinate product cone, hence an orthant and a simplex section.

In revisiting prior work, it distinguished its cone normalization from the covariance and Hessian ideas it had examined in Klartag's work [38]. It returned to complex-strip nonvanishing, Cauchy interpolation, and Gaussian-tail expansions in the scalar estimates.

43 Scalar positivity and the scope of the cone reformulation

The assistant audited the scalar estimates by separating short segments, tails, and middle widths. It checked the circle averages, the weighted trapezoid kernel, concentration bounds, and errors in the weighted primitives. In the middle range it retained the variance of q : a bound on $|q|$, together with a lower bound on the averaging density, gave a cubic lower bound for that variance in terms of end-point displacement. After accounting for the residual squares, it reduced the remaining inequalities to positivity of two polynomials, for which the Bernstein basis gives a sufficient criterion:

VERBATIM EXCERPT

| if $(T(l+d u)=\sum c_i u^i)$, then representation $(\sum_{r=0}^n b_r \binom{n}{r} u^r (1-u)^{n-r})$,
 $(b_r=\sum_{i \leq r} \binom{r}{i} c_i)$. all $(b_r > 0)$ ensures.

The geometric audit also questioned whether the cone statement had inadvertently strengthened the original problem. The assistant concluded:

VERBATIM EXCERPT

| Cone Laplace inequality exactly equivalent for hyperplane sections, not all log-concave.

With $m = n + 1$, it checked that the cone over $K - z_0$, evaluated at $U = (0, m)$ and $V = (0, 1)$, gives the Laplace-product factor $(n!)^2/m^m$. It revisited arbitrary nonsmooth cones through Lipschitz projection maps and Gaussian smoothing, rederived the covariance identities, and checked that squaring the matrix-square-root update yields the symmetrized Riccati equation without a commutation assumption. The assistant began another derivation of the bias construction and Brouwer normalization, reaching the inward-pointing estimates for eigenvalues of the coordinate-change matrix.

44 The orthant model and directional Gaussian tails

The assistant continued checking the proposed cone normalization, beginning with an inward-pointing estimate for the matrix parameter. It used this estimate in a Brouwer argument to impose both mean balance and covariance feedback. It then rederived the entropy comparison: integrated projection derivatives supplied Jacobians, a logarithmic determinant calculation supplied a surplus, and Gaussian smoothing and truncation preceded change of variables. Decomposing the centered layer field as $A = L + R$, it checked the Gaussian covariance identities, parity bounds, and commutator estimates intended to control the residual and the noncommuting covariance correction. Its equality argument proceeded from vanishing layer errors to commuting linear coefficients, coordinate thresholds, and an orthant after a linear transformation.

The orthant became an explicit consistency check and then an illustration in the draft. For $C = \mathbb{R}_+^m$, all-ones U , and covariance correction $T = 0$, the assistant obtained

$$\tilde{\zeta}_z = zU, \quad P_z = \text{diag}(1_{\{-G_i \leq z\}}), \quad A = L = \text{diag}(-G_i).$$

Integration of the transport weights gave coordinates $-\log p(-G_i)$ and $-\log p(G_i)$, where p is the standard Gaussian distribution function. It emphasized that general projection layers were not assumed to form a nested family of projections.

The tail estimates require different arguments at the two ends. For $s \geq 0$, the assistant wrote

$$a(-s) = \phi(s) \int_0^\infty u e^{-su - u^2/2} du \asymp \frac{\phi(s)}{(1+s)^2}.$$

This controlled the negative tail of P_z and X_z . At the positive end it instead used the bias deep inside the cone to control $I - P_z$ and Y_z . The vanishing statement concerns convergence as $z \rightarrow -\infty$.

45 Separability, variance recovery, and profile growth

The assistant reexamined its proposed general Mahler argument from the cone reduction through the scalar estimates.

It checked the biased Gaussian projection construction, the simultaneous choice of coordinates and covariance, and the entropy surplus involving traces of $P_x(I - P_z)$. Recomputing Gaussian integration by parts and the decomposition $A = L + R$, it followed the covariance identities, parity estimates, and layer errors through the noncommuting matrix terms. It checked that the matrix square-root update solved the symmetrized Riccati equation and that the resulting commutator costs matched the scalar endpoint inequality.

For equality, the assistant reasoned that vanishing off-diagonal spectral defect forces the coefficients of L to commute. Vanishing layer defect would then make the projection derivative diagonal in a fixed basis almost everywhere. It expanded the final geometric step: Lipschitzness and vanishing cross derivatives make each component of Π_C depend only on its corresponding coordinate, so its image C is a product. Solidity and pointedness restrict the coordinate factors to half-lines; undoing the transformation and slicing the cone would give a simplex.

The scalar audit tracked interpolation errors, weighted primitives, and separate error bounds for short intervals and tails. In the bounded middle range it retained $V_q = \overline{q^2} - \bar{q}^2$. A lower bound on the averaging density and a Lipschitz bound for q gave

$$V_q/h^2 \geq \frac{v_h z^3}{3 \cdot .193}, \quad z = \frac{|q_+ - q_-|}{2h}, \quad v_h = h \coth h - h.$$

The assistant checked how this cubic gain paid for the endpoint-error terms, then reduced the remaining polynomial inequalities to Bernstein-coefficient inequalities. For larger widths it revisited angular signs and rounded error bounds.

It also replaced a verbal allocation of variance with an explicit residual inequality.

A later rereading found an overbroad growth statement:

VERBATIM EXCERPT

| Potential ambiguity: $\backslash(X=p/\backslash\phi)\backslash$ grows like $\backslash(e^{\{x^2/2\}}\backslash)$, not polynomial.

The assistant revised the statement to name the test profiles whose derivatives have polynomial growth, excluding the temporary ratios X, Y . It also clarified the cone's positive heights, fixed-coordinate matrix expectations, and affine invariance, then continued checking signs, convergence, and equality deductions in the assembled candidate argument.

46 Reexamining the entropy bound and its matrix dependencies

The assistant returned to its general Mahler candidate, reconstructing the cone normalization, the entropy inequality, and the scalar estimates. For the cone over $K - z_0$, its positive dual D , and $m = n + 1$, it checked that the vertical choices $U = (0, m)$, $V = (0, 1)$ give

$$\chi_C(V)\chi_D(U) = \frac{(n!)^2}{m^m} |K| |(K - z_0)^\circ|.$$

It rederived the existence and tails of the biased Gaussian projection field, then the simultaneous choice of coordinates and covariance $\Sigma = I + T$. In particular, it checked that squaring the matrix square-root relation produced the intended quadratic equation without assuming that its coefficients commute. It also reconstructed the log-Jacobian surplus and the smoothing argument for the entropy transport.

The assistant next tracked the diagonal jump in the mixed derivative of $a(\min(x, z))B(\max(x, z))$, the Gaussian resolvent identity, and the signs and factors in the quadratic identities. Spectral thresholds supplied a positive layer measure even though the projection derivatives were not assumed nested. It

checked the Hermite-degree estimates, completed squares, and commutator bounds used to absorb the residual terms. Following its candidate's equality argument, it reasoned that vanishing losses would force commuting coefficient matrices and coordinatewise projection, hence an orthant cone and a simplex section.

It then expanded the argument for log-determinant integrability and clarified the Gaussian spectral form domain and endpoint orientation. In particular, it clarified that the clipped product in the bound for B_1 was convex as a function of one varying argument, with the other quantity fixed.

47 Cone normalization and matrix cancellation

The assistant continued auditing its general-case argument through the cone inequality

$$\chi_C(V)\chi_D(U) \geq 1, \quad \langle U, V \rangle = m = n + 1.$$

It rederived the biased-projection identities, the normalization by a linear transformation and Gaussian covariance $\Sigma = I + T$, and the entropy comparison. Particular checks concerned noncommuting matrices, determinant limits, convergence of the layer integrals, and differentiation for arbitrary nonsmooth cones. The assistant repeatedly tested whether a sign, normalization, or interchange of integrals had been lost; its orthant model supplied a case in which the proposed inequalities should become equalities.

Writing the integrated projection field as $A = L + R$, it checked the separation into Gaussian degrees, the contact terms, and the two spectral estimates controlling commutators. Direct integration reproduced the constants $2 \log 2 - 1/3$ and $\log 2 + 1/4$. Gaussian integration by parts on the linear field L , together with the quadratic equation for T , was then used to cancel the covariance term and leave the proposed scalar segment inequality. The assistant followed the equality argument from vanishing losses to commuting coefficient matrices, diagonal projection derivatives, and an orthant after transformation:

VERBATIM EXCERPT

Equality from negative losses doesn't require proving uniqueness of fixed point Q,T; for any one, if product=1 then all inequalities saturate, support yields transformed cone orthant.

48 Explicit projection maps and scalar dependencies

The assistant expanded the analytic steps of its general-case argument. It wrote the smoothed maps explicitly as integrals of biased cone projections, with $Z = \Sigma^{1/2}G$ and $\Sigma = I + T$. It used strict monotonicity for injectivity, then change of variables and Jensen's inequality for the entropy estimate. Its checks separated convergence of the linear costs from monotone convergence of the positive Jacobian factors and examined integrability of the limiting logarithmic determinants.

For covariance feedback it revisited the deterministic matrix equation

$$\lambda T^2 + \mathbf{C} \circ T + \mathbf{K} = 0, \quad B \circ E = (BE + EB)/2.$$

It emphasized that expanding the square and comparing positive square roots did not require simultaneous diagonalization:

VERBATIM EXCERPT

None of these matrices need commute; update uses completing square with sym product, and square-root comparisons are in Loewner order.

It then rederived Gaussian covariance identities, Hermite estimates, spectral threshold comparisons, and the cancellation of the term involving $\lambda \operatorname{tr}(\mathbf{S}\mathbf{T}^2)$. It checked the pointwise identity $F = d + |u|^2 = c + 2kq$ while distinguishing it from a multiplicative rule for the Ornstein–Uhlenbeck operator.

In its equality argument, strictness of the scalar segment inequality forced the spectral pair measure onto the diagonal, hence commutation of the linear Gaussian coefficients. Vanishing layer error then made the cone projection coordinatewise. The assistant concluded that the transformed cone was an orthant and its bounded section a simplex. It also checked the normalization on a two-dimensional simplex, obtaining volume product $27/4$.

The subsequent review returned to scalar estimates for Gaussian tails and expansions in x^{-2} . Intermediate segments required the variance of the auxiliary profile q , a lower bound from endpoint separation, and positivity of Bernstein coefficients.

49 The general-body conclusion

The assistant ends the general-case attempt by checking the fixed-point normalization, Gaussian covariance identities, entropy Jacobians, and noncommuting trace products. It emphasizes that the coefficients of the linear Gaussian matrix field are deterministic expectations, although chosen jointly with the covariance correction. It checks orthant and simplicial-cone cases, then follows vanishing layer and spectral errors to an orthant and a simplex. It adds details about Gaussian smoothing, boundary integrability, and the Hermite covariance formula. Its proposed conclusion remains dependent on the uniform scalar inequalities used in the entropy comparison.

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